



MODULE ELEVEN: ENDOCRINOLOGY AND REPRODUCTION

Year II Students

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The pelvis is divided into two parts by superior pelvic aperture. Upper/Greater / False pelvis mainly contains more of abdominal viscera where as true one contains pelvic viscera. The true pelvis has an inlet, an outlet, and a cavity. The false pelvis is of little clinical importance than true pelvis especially in female. sciatic notches are divided by the sacrotuberous and sacrospinous ligaments into the greater and lesser sciatic foramina. Piriformis: an important landmark which divide greater sciatic foramen into two.

Above the piriformis:

- Superior gluteal vessels & nerve and

Below the piriformis:

- Inferior gluteal vessels & nerve
- Sciatic nerve
- Posterior cutaneous nerve of thigh
- Pudendal nerve & Internal pudendal vessels
- Nerve to obturator internus
- Nerve to quadratus femoris

Structures passing through the lesser sciatic foramen

Entering:

- Pudendal nerve & Internal pudendal vessels and

Exiting:

- Tendon of obturator internus.

Because the sacrotuberous ligaments are strong and relatively inflexible, they should be considered to form part of the perimeter of the pelvic outlet. Pelvic outlet /inferior pelvic aperture/ is closed by pelvic floor.

Pelvic Wall

Wall of pelvis has three planes or layers:

- External plane or layer: is formed by gluteal muscles
- Intermediate plane is composed of: bones , ligaments & membranes
- Internal plane is formed by: muscles & fascia, blood vessels, lymphatic vessels & nerves and structures b/n peritoneum & fascial coverings of muscles.

Floor:- formed by peritoneum & pelvic diaphragm.

Obturator foramen is covered by obturator membrane except upper part with small obturator canal for the passage of obturator vessels & nerves

Lateral cup-like part of hip bone that articulates with head of femur (acetabulum) is formed by bodies of ilium & ischium & superior pubic ramus & is Surrounded by prominent bony margin called limbus acetabuli except inferior 1/3rd margin.

Inferiorly it has a notch called acetabular notch which is bridged by transverse acetabular ligament & is traversed by vessels supplying the joint.

Differences b/n Male & Female Hipbones		
Bony pelvis	Males	Females
General structure	Thick & heavy	Thin & lighter
Greater pelvis	Deep	Shallow
Lesser pelvis	Narrow, deep, tapering	Wide, shallow, cylindrical
Pelvic outlet	Comparatively small	Comparatively large
Pubic arch	Narrow ($< 70^\circ$)	Wide ($> 80^\circ$)
Obturator foramen	Round	Oval
Acetabulum	Large	Small
Great sciatic notch	Narrow ($\sim 70^\circ$)	Almost 90°
DA vs A-SP	Equal	Less
Ischial spine	Highly curved inward	Less curved inward
Preauricular sulci	absent	Infront of auricular surface

Pelvimetry

- It is measurement of pelvic diameter
- Important in gynaecology & obstetrics for delivery
- Measurable diameters could be internal & external and the internal is the more important one.

Types of Pelvis

Gynecoid pelvis:

- Found in 41 - 44% of females (z most common one).
- Typical female pelvis.

Android pelvis:

- Found in 23-32% of the females & is male like pelvis.
- Funnel-shaped pelvis / the pelvic inlet is heart-shaped.

Antropoid pelvis:

- Found in 23-28% of females.
- Ape like pelvis- long, narrow & oval in shape.
- Pelvic inlet has long anteroposterior diameter & short transverse diameter unlike others.

Platypelloid (flat) pelvis:

- Found in 2-5% of the females.
- Wide pelvis, flattened at pelvic inlet.
- Has wide subpubic arch as a conclusion it seems like gynecoid but delivery is difficult in this type of pelvis.

Digestive Organs in the Pelvis-Perineum

Are Rectum and Anal canal

Rectum

Has no sacculations, epiploic appendages and teniae coli. When rectum is viewed anteriorly, has 3 flexures. Flexures are formed in relation to 3 transverse foldings (Plicae transverse recti). The Middle rectal fold (valve) is the largest of the three & lie on the right side 7cm above the anus at the level of Douglas pouch. Help to separate solid waste product to gaseous.

Peritoneum coverings of rectum include:

- Superior third:- anterior & lateral surfaces covered
- Middle third:- only anterior surface is covered
- Inferior third:- no surface is covered.

Rectum has rectovesical or rectouterine pouch and pararectal fossae

- **Anterior Relations of the Rectum in males:** from fundus of urinary bladder to prostate gland.
- **Anterior Relations of the Rectum in females:** uterus and upper part of vagina.

Arterial supply of the Rectum

- Superior rectal artery,
- Middle rectal artery,
- Inferior rectal artery and
- Median sacral artery.

From these, superior rectal artery connects both systemic & portal system. So z cause for rectal hemorrhoid.

Anal Canal

Extend from superior portion of pelvic diaphragm to the anus. The canal is surrounded by internal (Involuntary) and external anal sphincters.

Blood Supply of Anal Canal; Arteries:- similar to the rectum.

There are two types of hemorrhoids.

Internal hemorrhoids; due to dilation of corpora cavernosa and lies above z anal sphincter

- bleeding has light red color because of no mixture of arterial & venous blood.

External hemorrhoids; is due to varicose of cutaneous veins & bleeding from this site is usually dark red

- found below anal sphincter under perineal skin .

MALE AND FEMALE UROGENITAL SYSTEM

URETER-thickest portion is near the entrance of urinary bladder ~ 10 mm & most stone occlusion occurs on it b/c it is z most curved part.

Intensive Care should be taken when there is uterus hysterectomy.

THE URINARY BLADDER

connected to the umbilicus by median umbilical ligament. Pubovesical ligaments extend from the bladder neck to the inferior aspect of the pubic bone. urinary bladder infection is called cystitis.

URETHRA- female ureters are more affected than male b/c of different reasons.

1. it is short
 2. more nearer to rectum
 3. is wider
 4. have no toxic substances like zinc that kill microbes
- any instrument that pass through external urethral orifice can easily reach to the bladder.

MALE GENITAL ORGANS

Testis-Coverings of the Testis; Scrotum, Tunica dartos, External spermatic fascia, Cremaster muscle, Internal spermatic fascia, Tunica vaginalis respectively from outer to inner.

Between the two layers of tunica vaginalis is scrotal cavity which contains a serous fluid. Excess fluid accumulation results in hydrocele.

Epididymis-Help to store spermatozoa until full maturation. Epithelial lining has two types of cells: columnar cells and basal cells

☐ Columnar cells contain non-motile cilia called stereocilia

Ductus Deferens-Ends by joining the duct of seminal vesicles (to form ejaculatory duct)

The Penis-

Smegma:- is sebaceous material secreted by sebaceous glands on the corona glandis & neck of penis these glands that secrete smegma are called prepuce glands.

Both parasympathetic & sympathetic work synergistically on it (exceptional).

Semen Consist of spermatozoa & additives from accessory reproductive glands, Come from seminal vesicle ~ 60%.

Epispadia:- urethra opens in the dorsum of the penis

Hypospadia:- urethra opens in the under surface the penis

Phimosis:- non retractable tightly fitted prepuce over the glans penis

Paraphimosis (capitration):- painful constriction of the glans penis by phimotic foreskin

Detumescence:- return of penis to flaccid state after orgasm because of sympathetic stimulation

Micropenis:- boys born with small penis (<2cm) but rare

Impotence:- failure of erection because of psychic, hormonal or neural problems

Priapism:- abnormal painful erection of penis for long time because of neural lesion

FEMALE GENITAL ORGANS

The Ovaries-On the medial portion is the hilum, for entrance of ovarian vessels & nerves. Ovarian Ligament attach uterine end to tubal angle on the uterus & contain ovarian branch of uterine artery

The Uterine Tube-Each tube opens into two ends:

1. proximal end:- to cornu or horn of uterus
2. distal end:- to peritoneal cavity near the ovary.

Ampulla Longest & widest portion of the uterine tube & is z Main site of fertilization .

The Uterus-Uterus usually bent anteriorly (anteflexed) b/n cervix & body with about 170° The entire uterus is inclined anteriorly (anteverted) from the vagina with 90°. Cervix has 2 types of epithelium.

Round ligament & ovarian ligament corresponds together to the gubernaculum of testes in males.

Arterial Supply of the Uterus is from both uterine arteries & small part from ovarian arteries.

Prolaps of uterus:- protrusion of uterus through the vaginal opening mainly after menopause & on multiparous patients.

Vagina-originated from 2 sources.

Episiotomy:- incision on external female genitalia

- done to facilitate delivery
- could be median or mediolateral
- both majora & minora are incised

PELVIC FLOOR AND PERINEUM

Superficial perineal space (pouch) - Lie b/n deep perineal fascia & perineal membrane

Contents of superficial perineal pouch

- Root of penis (two crura & one bulb)
- Bulb of the vestibule
- Crura of the clitoris
- Muscles:- superficial transverse perineal, ischioanal and bulbospongiosus
- Arteries:- posterior scrotal artery, transverse perineal artery and artery of penis
- Nerves:- posterior scrotal nerve, nerve to bulb and muscular branch of perineal nerve
- Glands & ducts:- greater vestibular gland in females and duct of bulbourethral gland in males

Deep Perineal Space (pouch) - Lie deep to the perineal membrane.

□ Contents of the deep perineal pouch

- Membranous urethra
- Muscles:- Sphincter urethra
- Deep transverse perineal

- Vessels:- Internal pudendal vessels
- Vessels of the bulb
- Nerves:- Dorsal nerve of penis
- Dorsal nerve of clitoris
- Glands:- bulbourethral glands in male

HISTOLOGY: Functional histology of male reproductive system

- **Testis**
- **Genital duct**
- **Accessory gland**
- **Penis**

Testis; male gonads where sperm and hormone are synthesized and released for their purpose of generation of perpetuation and secondary sexual characteristics.

Found in scrotum which is made of superficial fascia with sweat gland but no fat and smooth muscle called dartos and both have role in thermoregulation (optimum temperature for spermatogenesis-35°C).

Structures in testis from outside to inside are:

External spermatic fascia---EO

Cremasteric fascia and muscle---IO

Internal spermatic fascia--TF

} continuation of spermatic cord

Tunica vaginalis- peritoneal sac which consists of parietal and visceral layer where hydrocele, hematocele, e.t.c occurs

Tunica albuginea

Mediastinum testis – consists of seminiferous tubule (1-4 ST in each 250 pyramidal lobule)

Seminiferous tubule consist of tunica propira (fibroblast + **myoblast** (its contraction for propulsion of sperm) and germinal epithelium (seminiferous epithelium) where we find sertoli (supporting cell) and spermatogenic lineage cells (4-8 layer).

Spermatogenesis;

- Spermatocytogenesis; spermatogonia –primary spermatocyte
- Meiosis; primary spermatocyte ---secondary spermatocyte-spermatid
- Spermiogenesis; spermatid-- golgi phase—acrosomal phase—maturation phase--spermatozoa

Supporting (sertoli cells); elongated pyramidal cells with base adhering to basal lamina and apical end extending to the lumen. Give lateral process and form tight and gap junction – a basis **for blood testis barrier** which consist of basal (spermatogonia which have free blood access) and adluminal compartment

(which **immunoprivileged site** and contain spermatocyte and spermatid). They give physical, nutritional and metabolic support for immunoprivileged site.

Genital duct;

Duct	Epithelium	Connective Tissue	Muscle Layers	Function
Tubuli recti	Sertoli cells in proximal half; simple cuboidal epithelium in distal half	Loose	No smooth muscle	Conduct spermatozoa from seminiferous tubules to rete testis
Rete testis	Simple cuboidal epithelium	Vascular	No smooth muscle	Conduct spermatozoa from tubuli recti to ductuli efferentes
Ductuli efferentes	Regions of ciliated columnar cells alternating with unciliated cuboidal cells	Thin, loose	Thin layer of smooth muscle cells in circles	Conduct spermatozoa from rete testis to epididymis
Epididymis	Pseudostratified epithelium composed of tall principal cells with stereocilia, short basal cells	Thin, loose	Layer of smooth muscle cells in circles	Conduct spermatozoa from ductuli efferentes to ductus deferens
Ductus (vas) deferens	Pseudostratified stereociliated columnar epithelium	Loose, fibroelastic	Thick three-layer smooth muscle coats: inner, outer longitudinal; middle circular	Deliver spermatozoa from tail of epididymis to ejaculatory duct
Ejaculatory duct	Simple columnar epithelium	Subepithelial; thrown into folds, giving lumen irregular appearance	No smooth muscle	Deliver spermatozoa, seminal fluid to prostatic urethra at colliculus seminalis

Accessory genital gland;

Seminal vesicle; Epithelium of the seminal vesicles is pseudostratified columnar, with a height that varies with testosterone levels; it lines the extensively folded mucosa. secretes yellow, viscous fluid containing substances that activate sperm (e.g., fructose); this fluid constitutes about 70% of the human ejaculate.

Prostate; The prostate gland **surrounds the urethra (BPH)** as it exits the urinary bladder. The epithelium of the prostate gland is simple or pseudostratified columnar and lines the individual glands that constitute the prostate. Corpora amylacea are prostatic concretions, composed of glycoprotein, which may become calcified; their numbers increase with age.

Bulbourethral ; Bulbourethral glands are adjacent to the membranous urethra. They are lined by a simple cuboidal or columnar epithelium.

Penis ; corpora cavernosa and spongiosum (erectile tissue) and connective tissue make up the organ.

Histology: Female reproductive system

- The female reproductive system consists of the paired ovaries and oviducts as well as single uterus, vagina and external genitalia.
- Ovaries
- Each ovary is subdivided into a cortex and a medulla, which are not sharply delineated. It is covered by germinal epithelium and tunica albuginea
- Germinal epithelium is simple cuboidal epithelium

- Tunica albuginea a capsule that is composed of a dense, irregular collagenous connective tissue and found deeper to germinal epithelium.
- Cortex consists of ovarian follicles in various stages of development and a connective tissue stroma containing cells that respond to hormones. The stroma separates from follicular cells by basal lamina.
- Primordial follicles: 1^{ry} oocyte enveloped by a single layer of flat follicular cells. 1^{ry} oocyte arrested in prophase of meiosis I during fetal life and remain in this stage until ovulation.
- 1^{ry} follicles: possess zona pellucida a layer of extracellular material between the oocyte and granulosa cells
 - Unilaminar: single layer of cuboidal follicular cells surrounding 1^{ry} oocyte
- Multilaminar: several layers of follicular cells (now these cells also known as granulosa cells) surrounding 1^{ry} oocyte. The stroma has two layers: inner cellular layer (theca interna) which manufacture androgens and an outer fibrous layer (theca externa) which provide nourishment to the theca interna.
- 2^{ry} (antral) follicles: They possess zona pellucida, antrum (a large cavity between granulosa cells), cumulus oophorus (narrow processes extend from the granulosa cells into the zona pellucida) and corona radiata (granulosa cells immediately around and linked to the oocyte released together with oocyte during ovulation). Theca interna and externa are also present.
- Graafian (mature) follicle: is simply the dominant 2^{ry} follicle. Causes atresia of other 2^{ry} follicles.
 - Atretic follicles are follicles (in various stages of maturation) that are undergoing degeneration
 - Corpus luteum is a larger temporary endocrine gland formed by reorganization of the granulosa cells and theca interna of the ovulated follicle. It is composed of granulosa lutein cells (modified granulosa cells) which manufacture most of the body's progesterone and convert androgens into estrogens and theca lutein cells (modified theca interna cells) which manufacture progesterone and androgens
 - corpus albicans is the remnant of the degenerated corpus luteum
- Ovarian medulla contains large blood vessels, lymphatic vessels, and nerve fibers in a loose connective tissue stroma. They also possess a small number of estrogen-secreting interstitial cells and a few androgen-secreting hilus cells.

Organ	Mucosa		Muscular	Serosa/ adventitia
Fallopian tube	Infundibulum	Epithelium: simple columnar , consists of peg cells (secretion) and ciliated (transport) cells, extensive longitudinal folds	inner circular and outer longitudinal layer of smooth muscle	Serosa (thin connective tissue layer plus simple squamous epithelium)
	Ampulla	Differ from the above only by Less folds		
	Isthmus			
	Isthmus			
Uterus	Body	Endometrium: epithelial lining (simple columnar containing secretory and ciliated cells) and gland (simple tubular) rich connective tissue stroma. It has 2 layers functional layer: sloughed during the menstrual cycle & supplied by spiral arteries basal layer (basalis) is the deeper layer that is preserved during menstruation supplied by straight arteries	Myometrium: composed of 4 layers. Inner and outer longitudinal layers and a thick middle (2) circular layer. Blood supplied by arcuate artery	Serosa: surfaces of the uterus bulging into the peritoneal cavity. Adventitia: retroperitoneal surfaces of the uterus
	Fundus			
	Cervix not participate in	Simple columnar (mucus-secreting) epithelium except for the inferior portion is covered by a stratified squamous	little smooth muscle and consists	

	menstruation	nonkeratinized epithelium. Branched cervical glands	mainly of dense connective tissue	
Vagina	Nothing but a tube lined by nonkeratinized stratified squamous epithelium		two layers of smooth muscle, inner circular and outer longitudinal	Adventitia

External genitalia

- Labia majora are fat laden folds of skin; hair and secretions of sebaceous glands and sweat glands are present on their external surfaces. The inner surface is like labia minora
- Labia minora are folds of skin that possess a core of highly vascular connective tissue containing elastic fibers. They lack hair follicles, but their dermis contains numerous sebaceous glands, which open directly onto the epithelial surface.
- Vestibule is the space between the two labia minora. Glands of Bartholin (mucus-secreting glands) and numerous smaller mucus-secreting glands around the urethra open into this space.
- Clitoris is composed of erectile bodies which terminate in the prepuce cover glans clitoris. It contains many sensory nerve fibers and specialized nerve endings (e.g., Meissner corpuscles and pacinian corpuscles)

Functional histology of Pituitary, Thyroid, Parathyroid and Adrenal glands

Histology of Pituitary gland

The pituitary gland lies below the hypothalamus, to which it is structurally and functionally connected. Because of its dual origin, the pituitary actually consists of two glands—the **posterior neurohypophysis** and the **anterior adenohypophysis**—united anatomically but with different functions.

The neurohypophysis retains many histologic features of brain tissue and consists of a large part, the pars nervosa, and the smaller infundibulum stalk attached to the hypothalamus at the median eminence.

Pars nervosa, terminals of the hypothalamohypophysial tracts where the neurosecretory vesicles are accumulated, along with mitochondria, a few microtubules and smooth endoplasmic reticulum, at terminals as well as from the sides of the axon making the **herring bodies**. These are released near the fenestrated capillaries.

The neurohypophysis, unlike the adenohypophysis, does not contain the cells that synthesize its two hormones. Also present are highly branched glial cells called **pituicytes** that resemble astrocytes and are the most abundant cell type in the posterior pituitary. **pituicytes** are stellate cells that resemble astrocytes with cytoplasmic processes that are connected by gap junctions with each other and often terminate in the perivascular space. There are also fibroblasts and mast cells in the pars nervosa.

The neurosecretory bodies contain membrane-enclosed granules with either oxytocin or ADH bound to 10-kDa carrier proteins called neurophysin I and II, respectively.

Hypothalamic neurons in the SON synthesize vasopressin/antidiuretic hormone (ADH) and those of the PVN synthesize oxytocin, both of which are stored in axonal dilations called Herring bodies before release to capillaries in the posterior pituitary.

The adenohypophysis, derived from the oral ectoderm, has three parts: a large pars distalis or anterior lobe; the pars tuberalis, which wraps around the infundibulum; and the thin pars intermedia adjacent to the posterior pars nervosa.

1. The pars distalis, which constitutes 75% of the hypophysis, is supported by a connective tissue capsule and framework. It consists of chromophil, chromophobe and folliculostellate cells.

Chromophils are parenchymal cells that stain intensely because of their hormone-containing secretory granules. They synthesize, store, and release several hormones. These cells are subdivided into acidophils and basophils.

■ **Acidophils**- bind acid dyes and often stain orange or red and are primarily somatotrophs producing somatotropin (growth hormone), or lactotrophs (or mammotrophs) producing prolactin (PRL).

■ **Basophils**, which bind basic dyes and typically stain blue, include **gonadotrophs** producing follicle-stimulating hormone (FSH) and luteinizing hormone (LH) and thyrotrophs making thyroid-stimulating hormone (TSH). A third type of basophil is the corticotroph, synthesizing proopiomelanocortin (POMC) that is cleaved by proteases to make adrenocorticotrophic hormone (ACTH) and β -lipotropic hormone (LPH).

Chromophobes are parenchymal cells that stain poorly. It appears as small cells under the light microscope; the cells lack (or have only a few) secretory granules and are arranged close to one another in clusters.

Folliculostellate cells are numerous in the pars distalis and lie between the chromophils and chromophobes. They possess long processes that form gap junctions with processes of other folliculostellate cells. They surround hormone-producing cells and their function may be transmission of signals and physical support.

2. The pars intermedia lies between the pars distalis and pars nervosa. It contains many colloid-containing cysts (Rathke cysts) that are lined by cuboidal cells. It also possesses basophilic cells, which sometimes extend into the pars nervosa. These cells secrete (POMC), which is cleaved to form melanocyte-stimulating hormone (MSH). In humans, MSH acts in various ways to modulate inflammatory responses throughout the body, and it may play a role in controlling stores of body fat.

3. The pars tuberalis surrounds the cranial part of the infundibulum (hypophyseal stalk). It is composed of cuboidal basophilic cells, arranged in cords along an abundant capillary network. Its cells may secrete FSH and LH, but this has not been confirmed.

Vascularization of the pituitary gland

1. Arterial supply is from two pairs of blood vessels derived from the internal carotid artery.

a. The right and left superior hypophyseal arteries serve the pars tuberalis, infundibulum, and median eminence.

b. The right and left inferior hypophyseal arteries serve mostly the pars nervosa

2. venous system-drain the cavernous sinuses via several channels

Histology of thyroid gland

The thyroid gland is composed of two lobes connected by an isthmus. It has a capsule made by the deep cervical fascia and sends septa, which subdivide the thyroid gland into lobules containing follicles. These septa also serve as conduits for blood vessels, lymphatic vessels, and nerves.

Thyroid follicles are -spherical structures filled with colloid, a viscous gel consisting mostly of iodinated thyroglobulin. Surrounding the colloid within each follicle is a single layer of epithelial cells, called **follicular cells**. In addition, one or more **parafollicular cells**, occasionally lie sandwiched between the follicular cells. Both of these parenchymal cell types rest upon the basal lamina surrounding the follicle, which separates them from the abundant network of fenestrated capillaries in the connective tissue.

Function. Thyroid follicles synthesize, store, and release thyroid hormones.

1. Follicular cells are basophilic cells which are normally cuboidal, but they become columnar when stimulated and squamous when inactive. They possess a distended rough endoplasmic reticulum (RER) with many ribosome-free regions, a supranuclear Golgi complex, many lysosomes, and rod-shaped mitochondria. They synthesize and release T₃ and T₄.

2. Parafollicular cells are also called clear (C) cells because they stain less intensely than thyroid follicular cells. They are present singly or in small clusters of cells between the follicular cells and basal lamina. These cells belong to the population of DNEC cells, also known as APUD cells. They possess elongated mitochondria, substantial amounts of RER, a well-developed Golgi complex, and many membrane-bound dense secretory granules. They synthesize and release calcitonin.

Histology of the parathyroid gland

The parathyroid glands are four small glands that lie on the posterior surface of the thyroid gland, embedded in its connective tissue capsule. They are supported by septa from the capsule, which penetrate each gland and also convey blood vessels into its interior. They become infiltrated with fat cells in older persons, and the number of oxyphil cells also increases.

-Has three types of cells;

1. Chief (principal) cells-small and polygonal pale acidophilic cells with vesicular nuclei. Chief cells form anastomosing cords, surrounded by a rich, fenestrated capillary network.

These cells possess a central nucleus, a well-developed Golgi complex, abundant RER, small mitochondria, glycogen, and secretory granules of variable size.

They synthesize and secrete **parathyroid hormone** (PTH, or parathormone), which raises blood calcium levels. They maintain calcium level in the ECF by acting on three d/t target tissues: bone, kidneys and intestine (you will find the detail on this organ on physiology)

2. Oxyphil cells -are large, eosinophilic cells that are present singly or in small clusters within the parenchyma of the gland. Oxyphil cells possess many large, elongated mitochondria, a poorly developed Golgi complex, and only a limited amount of RER. They begin to appear after 10 years and increase with age. Their function is not known.

These are much larger than the principal cells and are characterized by very acidophilic

cytoplasm filled with abnormally shaped mitochondria.

3.transitional cells-with smaller nucleus that stain deeply than the others.

Histology of the adrenal gland

Adrenal glands lie embedded in fat at the superior pole of each kidney. The adrenal glands are invested by their own collagenous capsule.

Each gland has two concentric regions: a yellowish adrenal cortex and a reddish brown central adrenal medulla.

1.The adrenal cortex contains parenchymal cells that synthesize and secrete but do not store various steroid hormones. It is divided into three concentric histologically recognizable regions: the **zona glomerulosa, zona fasciculata, and zona reticularis**.

A. The zona glomerulosa, immediately inside the capsule and comprising about 15% of the cortex, consists of closely packed, rounded or arched cords of columnar or pyramidal cells with many capillaries.

Produce mineralocorticoid, mostly aldosterone in response to angiotensin II and to a lesser extent by ACTH.

B. The middle zona fasciculata, occupies 65% to 80% of the cortex and consists of long cords of large polyhedral cells, one or two cells thick, separated by fenestrated sinusoidal capillaries. cells contain many lipid droplets and (in tissue sections) appear so vacuolated that they are called spongiocytes. These cells also possess spherical mitochondria with tubular and vesicular cristae, SER, RER, lysosomes, and lipofuscin pigment granules.

These cells secrete **glucocorticoids**, especially **cortisol**, which affect carbohydrate metabolism by stimulating gluconeogenesis in many cells and glycogen synthesis in the liver. also have many effect on other organs.

C. The innermost zona reticularis comprises about 7% of the cortex and consists of smaller cells in a network of irregular cords interspersed with wide capillaries. cells are similar with those of zona reticularis; but have fewer lipid droplets, stain darkly acidophilic (since they contain fewer lipid droplets and large amount of lipofuscin pigment) and have large amount of lipofuscin pigment. they produce androgen, esp, DHEA.

2.adrenal medulla-is completely invested by the adrenal cortex. It contains two populations of parenchymal cells;

A.Chromaffin cells are large, polyhedral cells containing secretory granules that stain intensely with chromium salts (chromaffin reaction). Chromaffin cells are arranged in short, irregular cords surrounded by an extensive capillary network.

They are innervated by preganglionic sympathetic (cholinergic) fibers, making these cells analogous in function to postganglionic sympathetic neurons. They possess a well-developed Golgi complex, isolated regions of RE and numerous mitochondria. Catecholamine release occurs in response to intense emotional stimuli and is mediated by the preganglionic sympathetic fibers that innervate the chromaffin cells.

Two type of cells, 1. noradrenaline storing cells which make up 20% of the chromaffin cells and 2. adrenalin storing cells which make up 80% of the chromaffin cells are found within the chromaffin cell.

B.ganglion cells-fewer in number,most are sympathetic but a small number of parasympathetic neurons are also found. They innervate adrenal cortical cells and blood vessels.

Blood supply to the adrenal glands derived from the superior, middle, and inferior adrenal arteries, which form three groups of vessels: to the capsule, to parenchymal cells of the cortex, and directly to the medulla.

Development and congenital malformation of endocrine glands

1. Development of Hypophysis(pituitary gland):

Is attached to the hypothalamus by the pituitary stalk and consists of the anterior lobe (adenohypophysis) and the posterior lobe (neurohypophysis) which develops from two completely different parts:

1.1. Anterior lobe (adenohypophysis): develops from Rathke's pouch, which is an ectodermal diverticulum of the primitive oral cavity (stomodeum).

1.2. Posterior lobe (neurohypophysis): develops from the infundibulum,which is a neuroectodermal ventral evagination of the hypothalamus.

Hypophyseal defects:

Occasionally, Remnants of Rathke's pouch may give rise to a craniopharyngioma which is the most common supratentorial tumor occurring in childhood (i.e. most childhood tumor are inferior to the tentorium)and is the most common cause of hypopituitarism in children.

2. Development of thyroid gland:

The endodermal lining of the foregut forms the thyroid diverticulum. The thyroid diverticulum migrates caudally, passing ventral to the hyoid bone and laryngeal cartilages. During this migration, the thyroid remains connected to the tongue by the thyroglossal duct,which later is obliterated. The site of the thyroglossal duct is indicated in the adult by the foramen cecum.

N.B. unlike the follicular cells which are derived from endoderm parafollicular cells are derived from neural crest cells.

Thyroid gland defects

A. Ectopic thyroid tissue:

Result from the abnormal migration of the gland from its embryonic position to its definitive adult location. Glandular tissue may be found anywhere along its migratory path

B. Thyroglossal duct cyst:

Occurs when parts of the thyroglossal duct persist and thereby form a cyst. It is most commonly located in the midline near the hyoid bone, but it may also be located at the base of the tongue, when it is then called a lingual cyst. Is one of the most frequent congenital anomalies in the neck.

3. Development of parathyroid gland:

Parathyroid glands are found in two groups i.e. Inferior and superior parathyroid glands derived from third and fourth pharyngeal pouch respectively.

Parathyroid gland defects

Ectopic parathyroid tissue - the abnormal migration of these glands from their embryonic position to their definitive adult location. Glandular tissue may be found anywhere along their migratory path.

4. Development of adrenal gland:

4.1. Cortex: forms from two episodes of **mesoderm** proliferation that occur between the root of the dorsal mesentery and the gonad. The first episode forms the inner fetal cortex and the second episode forms the outer adult cortex, by which mesoderm proliferation occurs at the periphery of the fetal cortex. During the fetal period and at birth, the suprarenal glands are very large due to the size of the fetal cortex and become smaller as the fetal cortex involutes rapidly during the first 2 weeks after birth and continues to involute during the first year of life. The zona glomerulosa and zona fasciculata of the adult cortex are present at birth, but the zona reticularis is not formed until age 3 years.

4.2. Medulla: forms when **neural crest** cells aggregate at the medial aspect of the fetal cortex and eventually become surrounded by the fetal and adult cortex. The neural crest cells differentiate into chromaffin cells, which stain yellow-brown with chromium salts.

Clinical Considerations of suprarenal gland:

A. Neuroblastoma:

Is a common extra cranial neoplasm containing primitive neuroblasts of neural crest origin. Occurs mainly in children and is found in extra adrenal sites usually along the sympathetic chain ganglia (60%) or within the adrenal medulla (40%). NB metastasizes widely to the bone marrow, bone, and lymph nodes.

B. Pheochromocytoma:

Is a relatively rare neoplasm that contains both epinephrine and norepinephrine. PH occurs mainly in adults 40–60 years old and is generally found in the region of the adrenal gland, but it may be found in extra suprarenal sites. PH is associated with persistent or paroxysmal hypertension, anxiety, tremor, profuse sweating, pallor, chest pain, and abdominal pain.

C. Congenital adrenal hyperplasia (CAH):

Caused most commonly by mutations in genes for enzymes involved in adrenocortical steroid biosynthesis (e.g., **21-hydroxylase deficiency**, 11 β -hydroxylase deficiency). 21-hydroxylase deficiency accounts for 90% of all cases. No synthesis of the cortisol or aldosterone, so that intermediates are funneled into androgen biosynthesis, thereby elevating androgen levels. CAH is characterized by masculinization of a female fetus (i.e., female pseudo-intersexuality), adrenal hyperplasia due to unopposed ACTH production, and hyponatremia ("salt-wasting") with accompanying dehydration and hyperkalemia due to aldosterone deficiency.

- The genotype of the embryo (46,XX or 46,XY) is established at fertilization.
- During weeks 1–6, the embryo remains in a sexually indifferent or undifferentiated stage.
- This means that genetically female embryos and genetically male embryos are phenotypically indistinguishable.
- During week 7, the indifferent embryo begins phenotypic sexual differentiation.
- By week 12, female or male characteristics of the external genitalia can be recognized.
- By week 20, phenotypic differentiation is complete.

Phenotypic sexual differentiation:

determined by the SRY gene and may result in individuals with a female phenotype, an intersex phenotype, or a male phenotype. The SRY gene on chromosome Yp11.3 encodes for a sex-determining region Y (also called testes-determining factor [TDF]). As the indifferent gonad develops into the testes, Leydig cells and Sertoli cells differentiate to produce testosterone and Müllerian-inhibiting factor (MIF), respectively. In the presence of TDF, testosterone, and MIF, the indifferent embryo will be directed to the male phenotype and in the absence of TDF, testosterone, and MIF, the indifferent embryo will be directed to the female phenotype.

Components of the indifferent embryo:

Includes gonads, genital ducts, and primordia of external genitalia. Phenotypic sexual differentiation occurs in a sequence beginning with the gonads, then the genital ducts, and finally the primordia of external genitalia.

1. Development of the tests:

The intermediate mesoderm forms a longitudinal elevation along the dorsal body wall, the urogenital ridge. The coelomic epithelium and underlying mesoderm of the urogenital ridge proliferate to form the gonadal ridge. Primary sex cords develop from the gonadal ridge and incorporate primordial germ cells (XY genotype), which migrate into the gonad from the wall of the yolk sac. The primary sex cords form the seminiferous cords, tubuli recti, and rete testes. Seminiferous cords consist of primordial germ cells and Sertoli cells, which secrete Müllerian-inhibiting factor (MIF). The mesoderm between the seminiferous cords gives rise to the interstitial (Leydig) cells, which secrete testosterone.

The primordial germ cells, sustentacular (Sertoli) cells, interstitial (Leydig) cells, and connective tissue stroma of the testes are derived from mesoderm. The seminiferous cords remain as solid cords until puberty, when they acquire a lumen and are then called seminiferous tubules.

Relative descent of the testes:

The testes originally develop within the abdomen but later undergo a relative descent into the scrotum as a result of disproportionate growth of the upper abdominal region away from the pelvic region. The gubernaculum is a band of fibrous tissue along the posterior wall that extends from the caudal pole of the testes to the scrotum. The peritoneum evaginates alongside the gubernaculum to form the processus vaginalis. Later in development, most of the processus vaginalis is obliterated except at its distal end, which remains as a peritoneal sac called the tunica vaginalis of the testes.

2. Development of male genital ducts:

A. Paramesonephric (Müllerian) ducts: develop as invaginations of the lateral surface of the urogenital ridge. The cranial portions run parallel to the mesonephric ducts and the caudal portions fuse in the midline to form the uterovaginal primordium. Under the influence of MIF, the cranial portions of the paramesonephric ducts and the uterovaginal primordium regress.

B. Mesonephric (Wolffian) ducts and tubules: The mesonephric ducts form the epididymis, ductus deferens, seminal vesicle, and ejaculatory duct.

A few mesonephric tubules in the region of the testes form the efferent ductules of the testes.

C. Vestigial remnants:

- ✓ Vestigial remnants of the paramesonephric duct called the appendix testis
- ✓ Vestigial remnants of the mesonephric duct called the appendix epididymis
- ✓ Vestigial remnants of mesonephric tubules called the paradidymis

3. Development of male external genitalia:

Proliferation of mesoderm around the cloacal membrane causes the overlying ectoderm to rise up so that three structures are visible externally: the phallus, urogenital folds, and labioscrotal swellings.

- phallus forms the penis (glans penis, corpora cavernosa penis, and corpus spongiosum penis).
- urogenital folds form the ventral aspect of the penis (i.e., penile raphe).
- labioscrotal swellings form the scrotum.

Clinical considerations of male genital organs:

- A. **Hypospadias:** occur when the urethral folds fail to fuse completely, resulting in the external urethral orifice opening onto the ventral surface of the penis. It is generally associated with a poorly developed penis that curves ventrally, known as chordee.
- B. **Epispadias:** occur when the external urethral orifice opens onto the dorsal surface of the penis. It is generally associated with exstrophy of the bladder.
- C. **Bilateral undescended testes (cryptorchidism):** occurs when both testes fail to descend into the scrotum. Descent of the testes is evident within 3 months after birth. Cryptorchidism results in sterility. The undescended testes may be found in the abdominal cavity or in the inguinal canal. Unilateral undescended testis may also occur.
- D. **Hydrocele of the testes:** occurs when a small patency of the processus vaginalis remains so that peritoneal fluid can flow into the processus vaginalis, which results in a fluid-filled cyst near the testes.
- E. **Congenital inguinal hernia:** occurs when a large patency of the processus vaginalis remains so that a loop of intestine may herniate into the scrotum or labia majora. It is most common in males and is generally associated with cryptorchidism.

Development and congenital malformation of female reproductive organs

1. Development of the ovary:

The intermediate mesoderm forms a longitudinal elevation along the dorsal body wall, the urogenital ridge. The coelomic epithelium and underlying mesoderm of the urogenital ridge

proliferate to form the gonadal ridge. Primary sex cords develop from the gonadal ridge and incorporate primordial germ cells (XX genotype), which migrate into the gonad from the wall of the yolk sac.

Eventually, Primary oocytes, simple squamous cells, and connective tissue stroma of the ovary are derived from mesoderm.

Relative descent of the ovaries:

Ovaries originally develop within the abdomen but later undergo a relative descent into the pelvis as a result of disproportionate growth of the upper abdominal region away from the pelvic region. The gubernaculum is a band of fibrous tissue along the posterior wall that extends from the medial pole of the ovary to the uterus at the junction of the uterine tubes, forming the ovarian ligament. The gubernaculum then continues into the labia majora, forming the round ligament of the uterus. The peritoneum evaginates alongside the gubernaculum to form the processus vaginalis, which is obliterated in the female later in development.

2. Development of female genital ducts:

A. Paramesonephric (Müllerian) ducts: The cranial portions of the paramesonephric ducts develop into the uterine tubes and the caudal portions of the paramesonephric ducts fuse in the midline to form the uterovaginal primordium and thereby bring together two peritoneal folds called the broad ligament. The uterovaginal primordium develops into the uterus, cervix, and superior one third of the vagina. The paramesonephric ducts project into the dorsal wall of the cloaca and induce the formation of the sinovaginal bulbs and the sinovaginal bulbs fuse to form the solid vaginal plate, which canalizes and develops into the inferior two thirds of the vagina.

B. Mesonephric (Wolffian) ducts and tubules: they degenerate in the female after formation of the metanephric kidney.

C. Vestigial remnants: is related to the formation of cysts:

- hydatid cyst of Morgagni from hydatid of Morgagni remnant of the paramesonephric duct.
- Kobelt's cyst from the appendix vesiculosa remnant of the mesonephric duct.
- A cyst of the epoophoron from the epoophoron remnant of the mesonephric tubules.
- A cyst of the paroophoron from the paroophoron, remnant of the mesonephric tubules.
- A Gartner's duct cyst from the duct of Gartner remnant of the mesonephric duct

3. Development of female external genitalia:

- ✓ phallus forms the clitoris (glans clitoris, corpora cavernosa clitoris, and vestibular bulbs).
- ✓ The urogenital folds form the labia minora.
- ✓ labioscrotal swellings form the labia majora and mons pubis.

Clinical considerations of female genital organs:

- A. Atresia of the vagina:** is a condition in which the vaginal lumen is blocked due to a failure of the vaginal plate to canalize and form a lumen.
- B. Unicornuate uterus anomalies:** occur when one paramesonephric duct fails to develop or incompletely develops.
- C. Didelphys (double uterus) anomalies:** occur when there is a complete lack of fusion of the paramesonephric ducts.
- D. Bicornuate uterus anomalies:** occur when there is partial fusion of the paramesonephric ducts

- E. Septate uterus anomalies:** occur when the medial walls of the caudal portion of the paramesonephric ducts partially or completely fail to resorb.
- F. Hymen variations :** include the following: Crescentic hymen, Annular hymen, Redundant hymen, Imperforate hymen, Cribriform hymen, Microperforate hymen, Septate hymen
- G. Intersexuality:** Because the early embryo goes through an indifferent stage, events may occur by which a fetus does not progress toward either of the two usual phenotypes but gets caught in an intermediate stage known as intersexuality. Intersexuality is classified according to the histological appearance of the gonad and ambiguous genitalia.
- 1. True intersexuality:** occurs when an individual has both ovarian and testicular tissue (ovotestes) histologically, ambiguous genitalia, and a 46,XX genotype.
 - 2. Female pseudo-intersexuality:** occurs when an individual has only ovarian tissue histologically and masculinization of the female external genitalia. These individuals have a 46, XX genotype. Associated with a condition in which the fetus produces an excess of androgens (e.g., congenital adrenal hyperplasia [CAH]).
 - 3. Male pseudo-intersexuality:** occurs when an individual has only testicular tissue histologically and various stages of stunted development of the male external genitalia. These individuals have a 46, XY genotype. Most often observed clinically in association with a condition in which the fetus produces a lack of androgens (and MIF). Caused most commonly by mutations in genes for androgen steroid biosynthesis (e.g., 5 α -reductase 2 deficiency or 17 β -hydroxysteroid dehydrogenase). Increased T: DHT ratio is diagnostic. The reduced levels of androgens lead to the feminization of a male fetus.
Clinical findings: underdevelopment of the penis, scrotum and prostate gland (deficient DHT). The epididymis, ductus deferens, seminal vesicle, and ejaculatory duct are normal (enough T)
- H. Complete androgen insensitivity (CAIS; or testicular feminization syndrome):** is an X-linked recessive genetic disorder caused by a loss-of-function mutation in the AR gene on chromosome Xq12, which encodes for the androgen receptor. The lack of androgen receptor function results in defective virilization in 46,XY males despite the presence of bilateral testes and normal testosterone production. CAIS individuals have the following characteristics:
- primary amenorrhea, little or no axillary or pubic hair, there are normal-appearing female external genitalia; the labia and clitoris are normal or slightly underdeveloped;
 - the vagina is either absent or short and blind ending
 - there is absence or near absence Müllerian structures in the urogenital tract (i.e., uterus, uterine tubes, cervix, superior third of the vagina)
 - breast development is that of a normal female
 - Patients are taller and heavier than normal females; psychosocial orientation is female; and maternal instincts are present.

Hormone receptor and regulation

Hormone receptors are broadly categorized in to two:

1. Cell surface receptors (Receptor in cell membrane):

Are receptors for Water-soluble hormones that transmit signals to affect metabolism and gene expression without themselves entering the cytoplasm. They often do so via second messenger systems that, in turn, activate protein kinases. Example: Insulin, Glucagon, Catecholamine

Are of three types:

1.1. Ion channel receptors:

1.2. G-protein-coupled receptors

1.3. Kinase-linked receptors

Pathway	G Protein	Enzyme	Second Messenger(s)	Protein Kinase	Examples
cAMP	G _s (G _i)	Adenyl cyclase	cAMP	Protein kinase A	Glucagon Epinephrine (β , α -2) Vasopressin (V2, ADH) kidney
PIP ₂	G _q	Phospholipase C	DAG, IP ₃ , Ca ²⁺	Protein kinase C	Vasopressin (V1, V3) vascular smooth muscle Epinephrine (α_1)
cGMP	None	Guanyl cyclase	cGMP	Protein kinase G	Atrial natriuretic factor (ANF) Nitric oxide (NO)
Insulin, growth factors	Monomeric p21 ^{ras}	—	—	Tyrosine kinase activity of receptor	Insulin Insulin-like growth factor (IGF) Platelet-derived growth factor (PDGF) Epidermal growth factor (EGF)

Table. Summary of signal transduction by water soluble hormone

2. Intracellular receptors: contain domains that bind the hormone and domains that bind to regulatory elements (i.e., hormone response elements [HRE]) on DNA that stimulate or inhibit the synthesis of messenger RNA (mRNA). Translation of this mRNA produces proteins that are responsible for the physiologic effects of the hormone. Steroid and thyroid hormones, 1,25-dihydroxycholecalciferol(1,25-DHC), and retinoic acid cross the cell membrane and bind to intracellular receptors.

Hormone receptor regulation:

Receptor regulation is an important part of endocrine function and this occurs through up or down-regulation of the number of receptors and by desensitization of the receptors.

- Increasing or decreasing receptor synthesis
- Internalization of membrane receptors after ligand binding
- Uncoupling of the receptor from its signal transduction pathway (desensitization).

Some hormones may regulate their own receptors (homologous regulation) such as GnRH on the pituitary gonadotrophs whilst other receptors are regulated by other hormones (heterologous regulation) e.g. estrogen regulating oxytocin receptors.

Mechanism of Regulation of Hormone Secretion

- ❖ The nervous and endocrine systems work together to regulate and integrate the activities of the various organs in the body.
- ❖ The nervous system controls through electrochemical impulse, so the response is very fast.
- ❖ The endocrine system is a chemical messenger, since the chemicals have to be synthesized; secreted and transported it could take time, so slower.
- ❖ The hypothalamus is the part of the body which is used as the connection between the endocrine and the nervous systems. Because the hypothalamus has both neural and endocrine tissues.
- ❖ 3 Mechanisms
 1. Neural: example → Adrenal medulla's secretion is under sympathetic nervous control
 2. Chemical: example → PTH is regulated by blood calcium level
 3. Hormonal: example → when one hormone regulates the other.

Negative feedback and Positive feedback

Negative feedback:

- Accounts for most of the feedback control mechanisms
- Ensures a proper level of activity of the hormone at a target tissue

Positive feedback

- LH secretion is controlled via +ve feedback until ovulation. After ovulation the feedback stops.

Hypothalamus- Pituitary axis

- Have both anatomic and functional relationship
- Communicate via the infundibulum (pituitary stalk) which contains nerve fibres and small blood vessels.
- Work together to regulate the overall activity of peripheral endocrine glands.
- Special neurons in the hypothalamus synthesise and secrete the hypothalamic releasing and inhibitory hormones that control the anterior pituitary.

- The hormones of hypothalamus are released to blood via the hypophyseal portal system to anterior pituitary.

Hormones of the Hypothalamus	Primary action on Anterior Pituitary
Thyrotropin-releasing hormone	Stimulates secretion of TSH by thyrotropes
Gonadotropin- releasing hormone	Stimulates secretion of FSH and LH by gonadotropes
Corticotropin- releasing hormone	Stimulates secretion of ACTH by corticotopes
Growth hormone- releasing hormone	Stimulates secretion of GH by somatotropes
Growth hormone inhibiting hormone (somatostatin)	Inhibits secretion of GH by somatotropes
Prolactin- inhibiting hormone (Dopamine)	Inhibits secretion of PRL by lactotropes

Functions of Hormones of Anterior and Posterior Pituitary Gland

The anterior pituitary consists of two parts

- Pars distalis
- Pars tuberalis (the larger portion)

Anterior Pituitary Hormones

1. Growth Hormone

- Protein hormone with 191 amino acids
- Transported bound to plasma proteins → Half-life : 20min (longer)

Protein metabolism

- Increase amino acid transport through cell membrane, decrease plasma amino acid content
- Increase transcription and translation in the cells
- Decrease catabolism of proteins

Fat metabolism

- Increase mobilization of fat
- Increase free fatty acid level
- Increase utilization of fatty acids
- Provide a non-CHO source of substrate for ATP generation

Carbohydrate metabolism

- Decrease peripheral utilization of glucose
- Increase deposition of glycogen in cells
- Decrease the uptake of glucose by cells and increase plasma glucose level hence Diabetogenic effect

On bones

- Increase bone growth IGF-I
- Stimulates protein synthesis
- Multiplication of chondrocytes and osteogenic cells at the epiphyseal cartilage
- Conversion of chondrocytes into osteogenic cells and formation of new bones
- Enhance intestinal absorption of Ca²⁺ and mineralization of bone matrix

- In adults human growth hormone and IGFs help maintain the mass of muscles and bones and promote healing of injuries and tissue repair

2. Thyroid-stimulating hormone (TSH)

- Stimulates the synthesis and secretion of the two thyroid hormones, T3 and T4 by the thyroid gland.

3. Follicle-stimulating Hormone (FSH)

In females

- Initiates development of oocytes
- Induces secretion of estrogens

In males

- Stimulates sperm production in the testes. Affects Sertoli cells

4. Luteinizing hormone (LH)

In females

- Stimulates secretion of estrogen and progesterone
- Triggers ovulation
- Stimulates formation of the corpus luteum

In males

- Stimulates secretion of testosterone.

5. Prolactin

- Stimulates development of breast tissue
- Is the principal hormone for lactogenesis
 - ❖ Secretion is stimulated by Pregnancy, estrogen, Nursing, breast manipulation, TRH, oxytocin.
 - ❖ Secretion is inhibited by dopamine (PIF), somatostatin, GABA

Excess prolactin

- Blocks the synthesis and release of GnRH, prevents ovulation in females
- Decreases testosterone secretion and sperm production in males
- Cause lactation unassociated with pregnancy (galactorrhea)

Prolactin deficiency

- Cause destruction of the anterior pituitary gland inability to lactate.

6. Adrenocorticotrophic hormone (ACTH)

- Controls the production and secretion of cortisol and other glucocorticoids by the cortex of the adrenal glands.

7. Melanocyte-stimulating Hormone (MSH)

- MSH is produced from the same precursor molecule as ACTH -POMC.
- Binds to Melanotropin receptor, MC-1 melanocyte membrane Activates release of melanin by activation of tyrosinase enzyme.
 - ❖ ACTH binds to MC-1 receptor

Posterior Pituitary Gland (Neurohypophysis)

- Hormones : ADH (vasopressin) and Oxytocin
- Synthesis and secretion → Supraoptic and Periventricular nuclei
- Neurophysins → Binding proteins which transport ADH and Oxytocin through hypothalamus hypophyseal tract

1. ADH (Vasopressin)

- Increase water permeability of the distal nephron
- Vasoconstriction
- Messenger cell contraction
- Stimulates ACTH secretion
- Influences behavior, learning and memory.

Mechanism of Action

- ❖ The primary actions are mediated through V1a and V2 receptor types.

V1a receptors mediate the → vasoconstrictive effects

V2 receptors → predominant renal receptors

Control of Secretion

- Decrease in ECF volume (The sensitivity to volume changes is low at small volume changes.)
- Increase in osmolar concentration in ECF (osmoreceptors located in the hypothalamus are sensitive to increase in ECF osmolality)

2. Oxytocin

- Stimulates contraction of the, uterine myometrium myoepithelial cells surrounding the ducts and alveoli of the mammary gland

Other functions

- Role in sexual arousal
- Reduce anxiety, fear, mood disorder, against stress
- Improves social behavioral disorder
- Improves cognitive functions, learning, memory
- Facilitate transport of sperm through female genital tract to fallopian tube by causing uterine contraction during sexual intercourse
- In males Facilitates release of sperm into urethra during ejaculation.

Disorders of secretion of pituitary gland hormones

1 hyper function

A, Adenoma

- The cells that are not undergoing neoplastic change will be compressed leading to decrease in secretion of the non-neoplastic cells and increase in secretion of the neoplastic cells.
- This cellular monomorphism and the absence of a significant reticulin network distinguish pituitary adenomas from non-neoplastic anterior pituitary parenchyma
- macroadenoma (> 1cm) the optic nerve can be compressed leading to bitemporal-hemianopsie (the eyes can't perceive the outer part of vision).

2 Loss of function

A, Null cell adenoma

- Neoplastic change can arise in cells that are not secretory (0 cell adenoma), compressing all types of secretory cells in the anterior pituitary.

B, Sheehan syndrome (postpartum necrosis of the anterior pituitary)

- Blood loss (during pregnancy) → hypotension/hypovolemic shock → ischemic necrosis of the anterior pituitary (because of the loss of a direct arterial supply).
- Also encountered in disseminated intravascular coagulation, sickle cell anemia, elevated intracranial pressure, traumatic injury, and shock
- The anterior pituitary has substantial reserve capacity; as a result, destruction of large amounts of the anterior pituitary (75% or greater) must occur before signs and symptoms of hypopituitarism develop.
- The posterior pituitary, because it receives its blood directly from arterial branches, is much less susceptible

Posterior lobe defects

- Hyper-function is rare, but exists (i.e. SIADH, syndrome of inappropriate ADH/vasopressin production).
- loss of function (low ADH/vasopressin) causes Central diabetes insipidus (polyuria & polydipsia). it can result from several causes, including head trauma, neoplasms, and inflammation of hypothalamus and pituitary, and from surgical procedures involving the hypothalamus or pituitary and spontaneously (idiopathic).

Thyroid gland Functions and disorders of secretions

Function of thyroid

A, Follicular cells secrete T3&T4

- act directly, synergistically & permissively

1, Growth & Development

- permissive to GH & aids in the synthesis of IGF which promotes linear bone growth by increasing cellular uptake of amino acids & protein synthesis
- Promotes differentiation and maturation of
- fetal skeletal and nervous tissue
- Improve memory and learning capacity.
- Enhance wakefulness alertness and responsiveness to various stimuli
- Increase the speed and amplitude of the peripheral nerve reflexes.

2, ↑ in BMR

- ↑ O₂ consumption & ATP production in cells that have receptors for thyroid hormones (except in Brain, Retina, spleen, Testis & Lung) by ↑ the activity of membrane bound Na/K ATPase
- Thermogenesis (calorigenic effect)

3, Carbohydrate metabolism

- Have mixed effect
- At low doses potentiates the effect of insulin

- At high doses potentiates catecholamines and increases mobilization of glucose ($\uparrow$ its intestinal absorption, $\uparrow$ blood glucose & $\uparrow$ glucose uptake & use)
- Enhance glucose uptake by the muscle and adipose tissues
- Act directly on enzyme system involved in CHO metabolism

4, lipid metabolism

- **Lipolytic** by its self & by activating Hormon sensitive lipase
- Promote mobilization of fat Increase plasma level of free fatty acid for metabolism (Depletion of body's fat store, Decrease in weight, decrease plasma level of cholesterol and other lipids)
- Lower circulating cholesterol levels due to increased formation of LDL receptors in the liver resulting in increased hepatic removal of cholesterol from the circulation.

5, GIT

- **Stimulatory unlike Catecholamines**
- Increase appetite and food intake.
- Increase both the rates of secretion of digestive juices and motility of the GIT

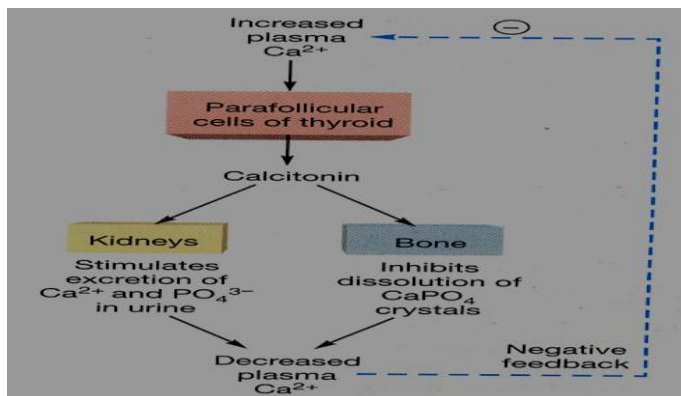
6, increase respiration

7, CVS

- $\uparrow$ CO, HR, SV, force of contraction & pulse pressure

8, Converts carotene into Vit A

B, Para follicular cells secrete calcitonin



Disorders of secretions

Hyper-function of the thyroid (Hyperthyroidism/thyrotoxicosis)

This is when the thyroid hormone secretion is elevated and this at a certain level can be toxic. This at times can be considered as a result of enlarged thyroid gland (functional goiter or neoplasm). But the case for

certain conditions like Graves' disease is not enlargement of the thyroid gland. Let's mention some conditions leading to hyperthyroidism

1, Graves' disease (most important cause of hyperthyroidism)

- An autoimmune disease where antibodies bind to the receptors for TSH.
- antibodies activate TSH independent activation of the receptors that can't be regulated by a feedback mechanism.
- Graves' disease the size of the follicles of the gland remain the same, but the number of follicular cells increase

2, Adenoma

- When it arises from the thyroid it can either be functional or non-functional (only focus on proliferation). But to cause hyperthyroidism it should be functional.
- The follicles of the adenoma are normal looking but their size is reduced compared to follicles of the normal thyroid tissue.

Malignancies of the Thyroid Gland

1, Follicular carcinoma –

- this closely resembles adenoma under the microscope with cytology but except for invading the capsule. It can also invade vessels with high affinity to blood vessels. This tumor may or may not produce the hormones.

2, Papillary carcinoma (finger like & most common 85%)

- has a great affinity for lymphatic vessels.
- Activation of the MAP kinase pathway is a feature of most papillary carcinoma
- the diagnosis of papillary carcinoma is based on nuclear features. The nuclei of papillary carcinoma cells contain very finely dispersed chromatin, which imparts an optically clear appearance, giving rise to the designation **ground glass** or "**Orphan Annie eye**" nuclei
- invaginations of the cytoplasm may give the appearance of intranuclear inclusions (hence the designation pseudoinclusions) in cross-sections.
- the neoplastic papillae have dense fibrovascular cores.
- Concentrically calcified structures termed **psammoma bodies** often are present within the papillae.

3, Anaplastic carcinoma –

- unlike the papillary or the follicular this one is not well differentiated. Because anaplastic carcinoma is poorly differentiated.
- The tendency to invade is greater for this tumor than the above two
- the prognosis is so low (weeks to months)

4, Poorly differentiated carcinoma – the other carcinomas are delimited but they can invade in to the delimiting capsule but this carcinoma is not well delimited in the first place.

5, Neuroendocrine carcinoma –

- this one arises from the parafollicular (C) cells. Unlike the other carcinomas of the thyroid this one is a medullary carcinoma and not a follicular carcinoma.

- good prognosis of 80% survival for 5 years. Microscopically it shows amyloidosis (red matter deposition).
- ✓ All the five tumors that we discussed (follicular carcinoma, papillary carcinoma, anaplastic carcinoma, poorly differentiated carcinoma and neuroendocrine carcinoma) are locally invasive and they could invade in to the trachea, causing respiratory insufficiency.

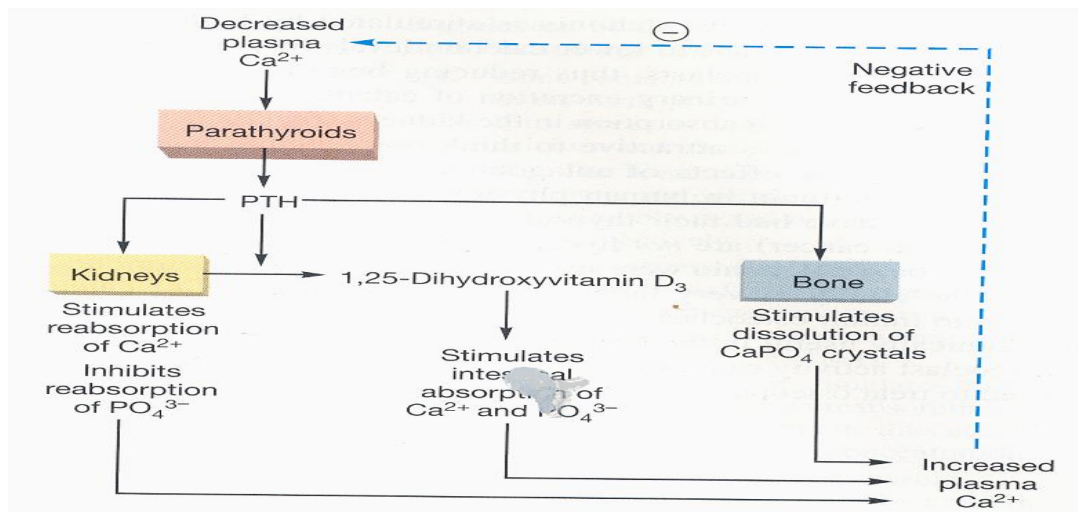
Hypothyroidism

1. hashimoto thyroiditis(main cause).

- an autoimmune disease where both the cellular and the humeral immunity are involved.
 - autoantibodies are secreted against thyroperoxidase enzyme (necessary in coupling the iodine to protein) and thyroglobulin.
2. It is known that there is no adenoma of the TSH secreting cells of the anterior pituitary. But an adenoma that arise from **non-secretory cells (0-cell adenoma)** can compress and atrophy the TSH secreting portion decreasing TSH secretion, subsequently decreasing T3 and T4 secretions.

Parathyroid gland Functions and disorders of secretion

Functions



- PTH enhances the release of calcium from Bones by Stimulating osteoblastic cells in the bone enhance expression of RANKL. RANKL Act on osteoclast precursors initiates their transformation to Osteoclasts.

Disorder

Hyper Function

1, Hyperplasia of the Parathyroid Gland

- It is the enlargement of the glands with the loss of the fat tissue of the gland, not neoplastic.

- It involves all the four parathyroid glands. It is caused due to persistent hypocalcaemia. The major cause of hypocalcaemia is renal insufficiency. Hyperplasia of the gland causes hyper-function. The renal insufficiency gives rise to hyperplasia of the gland leading to secondary hyperparathyroidism.

2, Adenoma

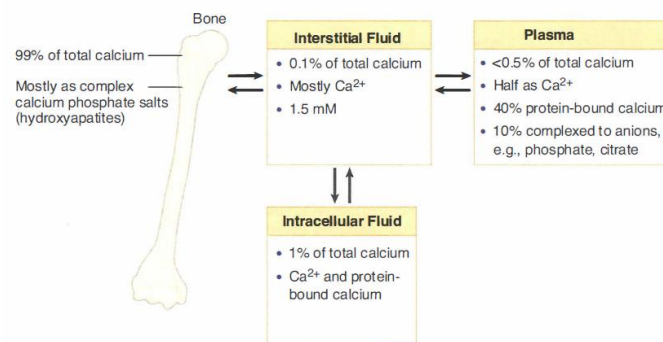
- leads to hyper-function. Adenoma gives primary hyperparathyroidism.
- Adenoma of the parathyroid hormone → elevated parathyroid hormone → elevated calcium level in blood → increase elimination in the urine → 20% get kidney stone
- ✓ majority of cases, primary hyperparathyroidism is caused by a sporadic parathyroid adenoma and, less commonly, by parathyroid hyperplasia.
- ✓ Parathyroid adenomas are solitary, while hyperplasia typically is a multiglandular process.
- ✓ Skeletal manifestations of hyperparathyroidism include bone resorption, osteitis fibrosa cystica, and brown tumors. Renal changes include nephrolithiasis (stones) and nephrocalcinosis.
- ✓ The clinical manifestations of hyperparathyroidism can be summarized as “painful bones, renal stones, abdominal groans, and psychic moans.”
- ✓ Secondary hyperparathyroidism most often is caused by renal failure, and the parathyroid glands are hyperplastic.
- ✓ Malignancies are the most important cause of symptomatic hypercalcemia, which results from osteolytic metastases or release of PTH-related protein from nonparathyroid tumors.

Hypo-function of the Parathyroid Gland

- accidentally removed during surgery on the thyroid gland or destroyed by an autoimmune mechanism.
- Congenital absence: This occurs in conjunction with thymic aplasia (Di George syndrome) and cardiac defects, secondary to deletions on chromosome 22
- Autoimmune hypoparathyroidism: This is a hereditary polyglandular deficiency syndrome arising from auto-antibodies to multiple endocrine organs (parathyroid, thyroid, adrenals, and pancreas).

Hormonal Control of Calcium and Phosphate

Calcium distribution in the body



Plasma calcium

Plasma calcium represents 50% ionized free, 40% attached to protein, 10% associated with anions such as phosphate and citrate.

- The free calcium is the physiologically active and precisely regulated form.

PARATHYROID HORMONE (PTH)

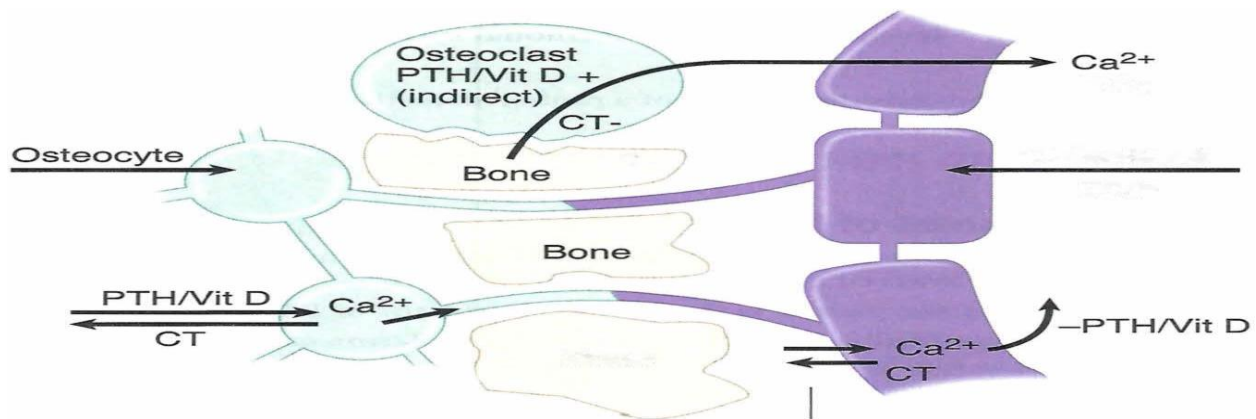
General Features of PTH and Its Actions:

- PTH is a peptide hormone released from the parathyroid glands in response to lowered plasma free Ca^{2+}
- Free Ca^{2+} in the plasma is the primary regulator of PTH.
- The negative feedback relationship between plasma calcium and PTH secretion is highly sigmoidal, with the steep portion of the curve representing the normal range of plasma free calcium.
- To sense the free calcium, the parathyroid cell depends upon high levels of expression of the calcium-sensing receptor (CaSR).
- In most cells exocytosis depends on a rise in intracellular free calcium. In the parathyroid gland, a fall in intracellular free calcium causes release.

Actions of PTH:

A decrease in the free calcium is the signal to increase PTH secretion and the function of PTH is to raise free calcium, which it does by several mechanisms.

- Increases Ca^{2+} reabsorption in distal tubule of the kidney
- Inhibits phosphate (Pi) reabsorption in proximal tubule of the kidney.
- Stimulates the 1-alpha-hydroxylase enzyme in kidney, converting inactive vitamin D to its active form.
- Causes bone resorption, releasing Ca^{2+} and Pi into the blood.



Bone resorption

The mechanisms of PTH-induced bone resorption are complex,

- Osteoblasts are cells that deposit bone and they contain receptors for PTH and vitamin D.
- Osteoclasts are cells that cause bone resorption.

- PTH binding to osteoblasts induces the synthesis of proteins, which ultimately causes differentiation of osteoclastic precursors to become functioning osteoclasts. Thus, circulating levels of PTH promote bone resorption.

CALCITONIN

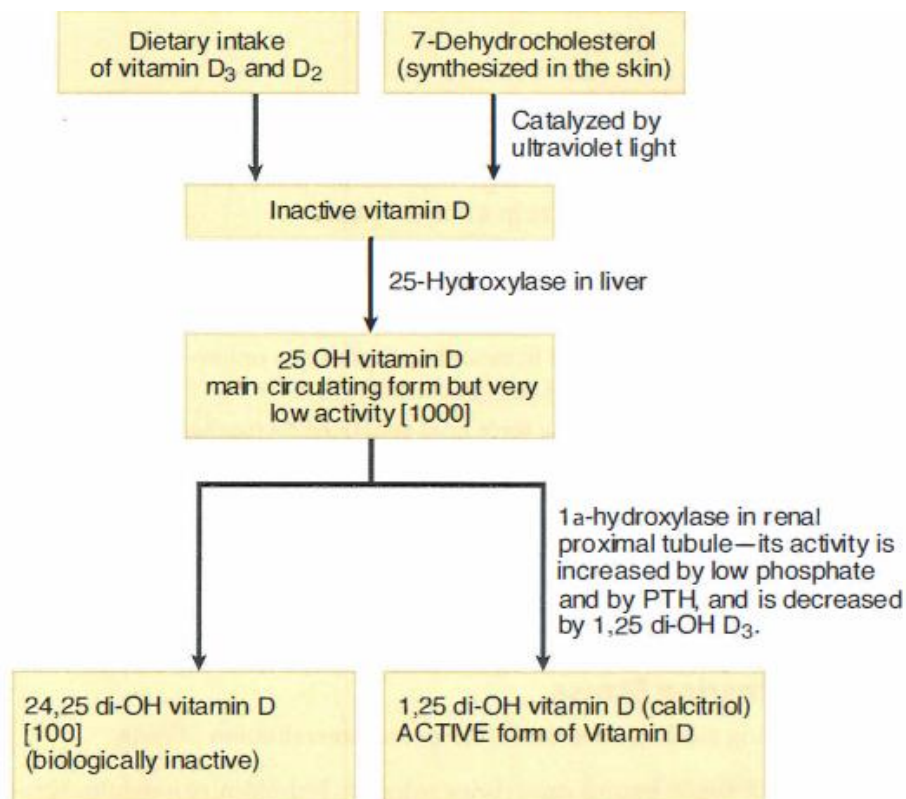
- Calcitonin (CT) is a peptide hormone secreted by the parafollicular cells (C cells) of the thyroid gland. It is released in response to elevated free calcium.
- Calcitonin lowers plasma calcium by decreasing the activity of osteoclasts, thus decreasing bone resorption. Calcitonin is useful in the treatment of Paget's disease, severe hypercalcemia, and osteoporosis.
- Calcitonin is not a major controller of Ca^{2+} in humans. Removing the thyroid (with the C cells) or excess of calcitonin via a C cell tumor (medullary carcinoma of the thyroid) has little impact on plasma calcium.
- No deficiency or excess disease has been described

ROLE OF VITAMIN D (CALCITRIOL) IN CALCIUM HOMEOSTASIS

Sources and Synthesis

Vitamin D₂ (ergocalciferol) is a vitamin but can functionally be considered a prohormone.

It is a normal dietary component. A slightly different form, vitamin D₃ (cholecalciferol), is synthesized in the skin. Its active form (1,25 di-OH D₃) is a hormone secreted by cells of the kidney's proximal tubule. The synthesis of 1,25 di-OH vitamin D (calcitriol)



The synthesis of calcitriol occurs sequentially in the skin - liver - kidney. The relative numbers of molecules of each of the hydroxylated forms of D present in the blood of a normal person are given in brackets. After its conversion to the 25 OH form in the liver, it can be stored in fat tissue. The serum levels of 25 OH vitamin D represent the best measure of the body stores of vitamin D when a deficiency is suspected. Most of the 25 OH form, which is the immediate precursor of 1,25 di-OH D, is converted to the inactive metabolite, 24,25 di-OH D. Ultraviolet (UV) light also evokes skin tanning, decreasing the penetration of UV light, and thus decreases the subsequent formation of D₃. This mechanism may prevent overproduction of D₃ in individuals exposed to large amounts of sunlight.

Actions of Calcitriol

Under normal conditions, vitamin D acts to raise plasma Ca²⁺ and phosphate. Thus, vitamin D promotes bone deposition. This is accomplished by

- Calcitriol increases the absorption of Ca²⁺ and phosphate by the intestinal mucosa by increasing the production of the Ca²⁺ -binding protein calbindin. The details of this process are poorly understood.
- The resulting high concentrations of Ca²⁺ and phosphate in the extracellular fluid exceed the solubility product, and precipitation of bone salts into bone matrix occurs.
- Calcitriol enhances PTH's action at the renal distal tubule.

At abnormally high activity levels calcitriol increases bone resorption and release of Ca²⁺ and phosphate from bone. Receptors for calcitriol are on the nuclear membranes of osteoblasts. Through communication from osteoblasts, activated osteoclasts carry out the bone resorption. Calcitriol requires the concurrent presence of PTH for its bone-resorbing action.

Adrenal gland hormones, Regulation of secretion Functions and Disorders of secretion

The **adrenal glands** (also known as **suprarenal glands**). Each adrenal consists of an outer cortex and inner medulla. The adrenal cortex and adrenal medulla are structurally and functionally different. The adrenal medulla secretes catecholamine hormones, which complement the sympathetic nervous system in the “fight-or-flight” reaction. The adrenal cortex secretes steroid hormones that participate in the regulation of mineral and energy balance.

The Adrenal Cortex

Has three distinct layers

1. Zona Glomerulosa

A thin layer of cells that lies just underneath the capsule constitutes about 15 percent of the adrenal cortex. These cells are the only ones in the adrenal gland capable of secreting Mineralocorticoids like significant amounts of *aldosterone* because they contain the enzyme *aldosterone synthase*. The secretion of these cells is controlled mainly by the extracellular fluid concentrations of *angiotensin II* and *potassium*, both of which stimulate aldosterone secretion.

2. The zona fasciculate

The middle and widest layer, constitutes about 75 percent of the adrenal cortex and secretes glucocorticoids like *cortisol* and *corticosterone*. The secretion of these cells is controlled in large part by the hypothalamic-pituitary axis via *adrenocorticotrophic hormone* (ACTH).

3. The *zona reticularis*

The deep layer of the cortex, secretes the adrenal androgens *dehydroepiandrosterone* (DHEA) and *androstenedione*. ACTH also regulates secretion of these cells, although other factors such as *cortical androgen-stimulating hormone*, released from the pituitary, may also be involved.

The Adrenal Medulla

The cells of the adrenal medulla secrete **epinephrine** and **norepinephrine** in an approximate ratio of 4 to 1. The adrenal medulla is at the centre of each adrenal gland. The chromaffin cells of the medulla are the body's main source of the catecholamine adrenaline and noradrenaline, released by the medulla. Approximately 20% noradrenaline (norepinephrine) and 80% adrenaline (epinephrine) are secreted here.

The effects of these catecholamine hormones are similar to those caused by stimulation of the sympathetic nervous system, except that the hormonal effect lasts about 10 times longer.

Function of Adrenal Hormone

A. Mineralocorticoids

Aldosterone is the most potent mineralocorticoid. **Renal and Circulatory Effects of Aldosterone**

Aldosterone increases reabsorption of sodium and simultaneously increases secretion of potassium by the renal tubular epithelial cells, especially in the *principal cells of the collecting tubules* and, to a lesser extent, in the distal tubules and collecting ducts. Therefore, aldosterone causes sodium to be conserved in the extracellular fluid while increasing potassium excretion in the urine. These actions help to increase the blood volume and pressure and to regulate blood electrolyte balance.

Excess aldosterone increases extracellular fluid volume and arterial pressure but has only a small effect on plasma sodium concentration, also causes hypokalemia and muscle weakness plus increases tubular hydrogen ion secretion and causes alkalosis. Too little aldosterone cause hyperkalemia and cardiac toxicity.

B. Glucocorticoids

The predominant glucocorticoid in humans is *cortisol* (*hydrocortisone*). Cortisol and other glucocorticoids have many effects on metabolism. As their name suggests, they increase the circulating level of glucose. This is the result of an increase in the mobilization of amino acids from protein and the stimulation of synthesis of glucose from these amino acids in the liver. In addition, they increase the levels of free fatty acids, which cells can use as an alternative to glucose to obtain energy. So we can say they promote lipolysis, and stimulate gluconeogenesis. Glucocorticoids also have effects not related to the regulation of blood sugar levels, including the suppression of the immune system and a potent anti-inflammatory effect. Cortisol reduces the capacity of osteoblasts to produce new bone tissue and decreases the absorption of calcium in the gastrointestinal tract.

Cortisol is not evenly released during the day – its concentrations in the blood are highest in the early morning and lowest in the evening as a result of the circadian rhythm of ACTH secretion. Cortisone is an

inactive product of the action of the enzyme 11β -HSD on cortisol. The reaction catalyzed by 11β -HSD is reversible, which means that it can turn administered cortisone into cortisol, the biologically active hormone.

C. Androgens

Cells in zona reticularis of the adrenal glands produce male sex hormones, or androgens, the most important of which is DHEA (*dehydroepiandrosterone*). In general, these hormones do not have an overall effect in the male body, and are converted to more potent androgens such as testosterone and DHT or to estrogens (female sex hormones) in the gonads, acting in this way as a metabolic intermediate.

D. Catecholamines

Adrenaline and noradrenaline are water-soluble compounds that have a structure made of a catechol group and an amine group. The adrenal glands are responsible for most of the adrenaline that circulates in the body, but only for a small amount of circulating noradrenaline. These hormones are released by the adrenal medulla, which contains a dense network of blood vessels. Adrenaline and noradrenaline act at adrenoreceptors throughout the body, with effects that include an increase in blood pressure and heart rate. The actions of adrenaline and noradrenaline are responsible for the fight or flight response, characterized by a quickening of breathing and heart rate, an increase in blood pressure, and constriction of blood vessels in many parts of the body.

The hormones from the adrenal medulla increase the cardiac output and heart rate, dilate coronary blood vessels, increase mental alertness, increase the respiratory rate, and elevate the metabolic rate. The adrenal medulla is innervated by preganglionic sympathetic axons and secretes its hormones whenever the sympathetic nervous system is activated during “fight or flight”. These sympathoadrenal effects are supported by the metabolic actions of epinephrine and norepinephrine: a rise in blood glucose due to stimulation of hepatic glycogenolysis (breakdown of glycogen) and a rise in blood fatty acids due to stimulation of lipolysis (breakdown of fat).

Regulation of Adrenal Hormone

Adrenal Medulla

The adrenal medulla is driven by the sympathetic nervous system via preganglionic fibers originating in the thoracic spinal cord, from vertebrae T5–T11. Because it is innervated by preganglionic nerve fibers, the adrenal medulla can be considered as a specialized sympathetic ganglion. Unlike other sympathetic ganglia, however, the adrenal medulla lacks distinct synapses and releases its secretions directly into the blood.

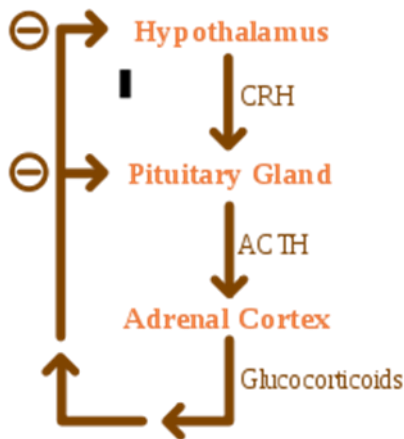
Regulation Zona glomerularis

- AgII : through DAG, protein kinase C
- Potassium ; Via voltage gated calcium channels
- ACTH : cAMP, protein kinase A
- Sensitivity of ZG to AgII and to low Na diet is decreased by a low K diet.

Regulation of zona fascicularis

Cortisol exerts negative feedback control on the very axis that stimulates its secretion, and it does so at the level of both the anterior pituitary and hypothalamus.

- Feedback to the Anterior Pituitary
- Feedback to the Hypothalamus



Regulation of Zona Reticularis

ACTH is the primary regulator of the zona reticularis. Both DHEA and androstenedione display the same diurnal rhythm as cortisol (DHEAS does not because of its long circulating half-life). Moreover, the zona reticularis shows the same atrophic changes as the zona fasciculata in conditions typified by little or no ACTH. However, other factors must regulate adrenal androgen function. Adrenarche occurs in the face of constant ACTH and cortisol levels, and the rise and decline of DHEAS is not associated with a similar pattern of ACTH or cortisol production. However, the other factors, whether extraadrenal or intraadrenal, remain unknown.

Disorders of secretion

An excessively high level of corticosteroids in the blood causes **Cushing's syndrome**. This may result from the over secretion of ACTH (usually by a tumor of the anterior pituitary) that overly stimulates the adrenal cortex to secrete corticosteroids, but it can also be produced by a tumor of the adrenal cortex that secretes excessive amounts of corticosteroids. Cushing's syndrome is characterized by changes in carbohydrate and protein metabolism, hyperglycemia, hypertension, and muscular weakness. Metabolic problems give the body a puffy appearance and can cause structural changes characterized as "buffalo hump" and "moon face."

Addison's disease is caused by inadequate secretion of both glucocorticoids and mineralocorticoids, which results in hypoglycemia, sodium and potassium imbalance, dehydration, hypotension, rapid weight loss, and generalized weakness. A person with this condition who is not treated with corticosteroids will eventually die because of severe electrolyte imbalance and dehydration. In the original description of the disease by Addison (1793–1860), he described a "characteristic discoloration of the skin . . . smoky appearance, or various tints or shades of deep amber or chestnut brown." This is caused by the very high secretion of ACTH, which at that concentration can stimulate melanocytes. High ACTH secretion results from inadequate negative feedback (due to low glucocorticoid secretion).

A tumor of the adrenal medulla is referred to as a **Pheochromocytoma**. This tumor causes hyper secretion of epinephrine and norepinephrine, which produces an effect similar to continuous sympathetic nerve stimulation. The symptoms of this condition are hypertension, elevated metabolism, hyperglycemia and

sugar in the urine, nervousness, digestive problems, and sweating. It does not take long for the body to become totally fatigued under these conditions, making the patient susceptible to other diseases.

Pancreatic gland Functions and Disorders of secretion

The endocrine portion of the pancreas consists of scattered clusters of cells called the **pancreatic islets** or **islets of Langerhans**. These endocrine structures are most common in the body and tail of the pancreas. The islets contain three major types of cells, *alpha*, *beta*, and *delta* cells.

Alpha cells: 25% of islets and produce glucagon.

Beta cells: 60% of the cells and produce insulin.

Delta cells: 10% of the islet and produce gastrin and somatostatin.

Insulin

- Peptide hormone consisting of two chains, an alpha and beta chains connected by two disulfide bridges.
- Acts on tissues that require insulin for effective uptake of glucose and amino acids especially in liver, skeletal muscle, adipose.
- Insulin does not regulate glucose transport into brain, RBC(diffuses in), lens of the eyes, mucosa of the small intestine and renal tubular brush borders
- Species Specificity.
- Insulin causes Potassium to shift into the cells thereby decreasing the extracellular K level.

Glucose transport into cells

- Major carriers include GLUT-1 and GLU-3
- GLUT-2 is found in the liver and pancreas.
- SGLT-1 is involved in the sodium- dependent secondary active transport of glucose by intestinal epithelium and renal tubular cells.
- GLUT-4 is regulated by insulin. It is present in skeletal muscle, cardiac muscle and adipose tissue.

Function of insulin

- Increases glycogen production (glucose storage) in the liver and muscle.
- Stimulates lipid synthesis from free fatty acids and triglycerides in adipose tissue.
- Also stimulates potassium uptake by cells (role in potassium homeostasis).
- ↑ glucose uptake in most cells (not in active muscle, only resting muscle)
- ↑ glucose use and storage
- ↑ protein synthesis

- ↑ fat synthesis
- Insulin like GH is essential to growth. Insulin and GH interact synergistically to promote growth.

Regulation of insulin secretion

Insulin secretion is increased by elevated blood glucose concentrations, gastrointestinal hormones (secretin, CCK), Parasympathetic (Acetylcholine, Glucagon), Serum free fatty acids, a rise in amino acids (arginine, lysine) and Beta adrenergic stimulation.

Insulin secretion is inhibited by catecholamines and somatostatin.

Glucagon

Glucagon, secreted by the alpha cells of the pancreatic islets, acts antagonistically to insulin—it promotes effects that raise the plasma glucose concentration. Glucagon secretion is stimulated by a fall in the plasma glucose concentration and insulin secretion that occurs when a person is fasting. Under these conditions, glucagon stimulates the liver to hydrolyze glycogen into glucose (a process called *glycogenolysis*), allowing the liver to secrete glucose into the blood. Glucagon, together with the glucocorticoid hormones, also stimulates *gluconeogenesis*—the conversion of noncarbohydrate molecules into glucose—to help raise the plasma glucose level during times of fasting. In addition, glucagon and other hormones promote other catabolic effects, including *lipolysis* (the hydrolysis of stored fat) and *ketogenesis* (the formation of ketone bodies from free fatty acids by the liver). These free fatty acids and ketone bodies serve as energy sources for cell respiration during times of fasting.

Regulation of glucagon

Secretion of glucagon is stimulated by:

- Hypoglycemia
- Epinephrine (via β_2 , α_2 , and α_1 adrenergic receptors)
- Arginine
- Alanine (often from muscle-derived pyruvate/glutamate transamination)
- Acetylcholine
- Cholecystokinin

Secretion of glucagon is inhibited by:

- Somatostatin
- Insulin (via GABA)
- PPAR γ /retinoid X receptor heterodimer.
- Increased free fatty acids and keto acids into the blood.
- Increased urea production

Somatostatin

The *delta cells* of the islets of Langerhans secrete the hormone *somatostatin*, a 14 amino acid polypeptide that has an extremely short half-life of only 3 minutes in the circulating blood. Almost all factors related to the ingestion of food stimulate somatostatin secretion. They include (1) increased blood glucose, (2) increased amino acids, (3) increased fatty acids, and (4) increased concentrations of several of the gastrointestinal hormones released from the upper gastrointestinal tract in response to food intake. In turn, somatostatin has multiple inhibitory effects as follows:

1. Somatostatin acts locally within the islets of Langerhans themselves to depress the secretion of both insulin and glucagon.
2. Somatostatin decreases the motility of the stomach, duodenum, and gallbladder.
3. Somatostatin decreases both secretion and absorption in the gastrointestinal tract.

Putting all this information together, it has been suggested that the principal role of somatostatin is to extend the period of time over which the food nutrients are assimilated into the blood. At the same time, the effect of somatostatin to depress insulin and glucagon secretion decreases the utilization of the absorbed nutrients by the tissues, thus preventing rapid exhaustion of the food and therefore making it available over a longer period of time.

It should also be recalled that somatostatin is the same chemical substance as *growth hormone inhibitory hormone*, which is secreted in the hypothalamus and suppresses anterior pituitary gland growth hormone secretion.

Diseases and Disorders of the Pancreas

Problems in the production or regulation of pancreatic hormones will cause complications related to blood sugar imbalance.

- **Type 1 diabetes**

Type 1 diabetes mellitus is because of destruction of the beta cells with an autoimmune mechanism (CD8⁺ T cells), so insulin production is reduced. For this autoimmune attack there can be a genetic susceptibility. Childhood viral infection can also lead trigger the immune mechanism. Type 1 diabetes is rare compared to type 2. It is worthy to know that the onset of type 1 diabetes mellitus is at an early age, childhood or puberty. So we can also call it juvenile diabetes mellitus. Although the type 1 is rare it is more severe in its clinical manifestations.

- **Type 2 diabetes**

It's due to insensitivity of the target organs to insulin. When the glucose uptake is blocked the beta cells will start producing insulin in excess and in the early stages this over production will be compensatory. But overtime the beta cells will be exhausted of producing such a large amount of insulin leading to the impairment of insulin secretion .People with type 2 diabetes may be able to produce insulin, but their bodies don't use it correctly. They might also be unable to produce enough insulin to handle the glucose in their body. Lifestyle choices, such as diet and exercise, play a major role in managing and preventing type 2 diabetes.

Other common diseases and disorders associated with the pancreas are:

- **Hyperglycemia:** This condition is caused by abnormally high blood glucose levels. It can be caused by overproduction of the hormone glucagon. To learn more, read our article about hyperglycemia.
- **Hypoglycemia:** Conversely, hypoglycemia is caused by low blood glucose levels. It is caused by a relative overproduction of insulin. To learn more, read our article about hypoglycemia.

Despite the fact that the great majority of pancreatic cells are devoted to digestive function, the endocrine cells play a major role in your overall health. By regulating your blood sugar levels, the pancreatic hormones are directly related to some of the most common diseases of today, including diabetes.

Pineal gland Regulation of secretion and Functions

The **pineal gland**, also known as the **pineal body**, **conarium** or **epiphysis cerebri**. The pineal gland is a tiny endocrine gland found in the brain. It's no bigger than a pea and can be found situated between the right and left hemispheres of your brain. Melatonin—the only hormone that the gland is known to produce and release. Melatonin helps maintain circadian rhythm and regulate reproductive hormones. The pineal gland often appears calcified in x-rays, which is usually due to fluoride, calcium, and phosphorus deposits that build up with age.

How does the retina transmit information about light-dark exposure to the pineal gland?

Light exposure to the retina → suprachiasmatic nucleus of the hypothalamus → descend to the spinal cord to and project into superior cervical ganglia → post-ganglionic neurons ascend back to the pineal gland.

Regulation of Melatonin

The precursor to melatonin is serotonin, a neurotransmitter that itself is derived from the amino acid tryptophan. Within the pineal gland, serotonin is acetylated and then methylated to yield melatonin.

Synthesis and secretion of melatonin is dramatically affected by light exposure to the eyes. The fundamental pattern observed is that serum concentrations of melatonin are low during the daylight hours, and increase to a peak during the dark. The duration of melatonin secretion each day is directly proportional to the length of the night.

The mechanism behind this pattern of secretion during the dark cycle is that activity of the rate-limiting enzyme in melatonin synthesis - serotonin N-acetyltransferase (NAT) - is low during daylight and peaks during the dark phase. Activity of the other enzyme involved in synthesis of melatonin from serotonin - the methyltransferase - does not show regulation by pattern of light exposure.

Two melatonin receptors have been identified from mammals

Mel1A and Mel1B that are differentially expressed in different tissues and probably participate in implementing differing biologic effects. These are G protein-coupled cell surface receptors. The highest density of receptors has been found in the suprachiasmatic nucleus of the hypothalamus, the anterior pituitary (predominantly pars tuberalis) and the retina. Receptors are also found in several other areas of the brain.

Biological effects of melatonin

1) Effect on reproductive function

The effect of melatonin on reproductive systems can be summarized by saying that it is anti-gonadotropic. In other words, melatonin inhibits the secretion of the gonadotropic hormones luteinizing hormone and follicular stimulating hormone from the anterior pituitary. Much of this inhibitory effect seems due to inhibition of gonadotropin-releasing hormone from the hypothalamus, which is necessary for secretion of the anterior pituitary hormones.

2) Effects on Sleep and Activity

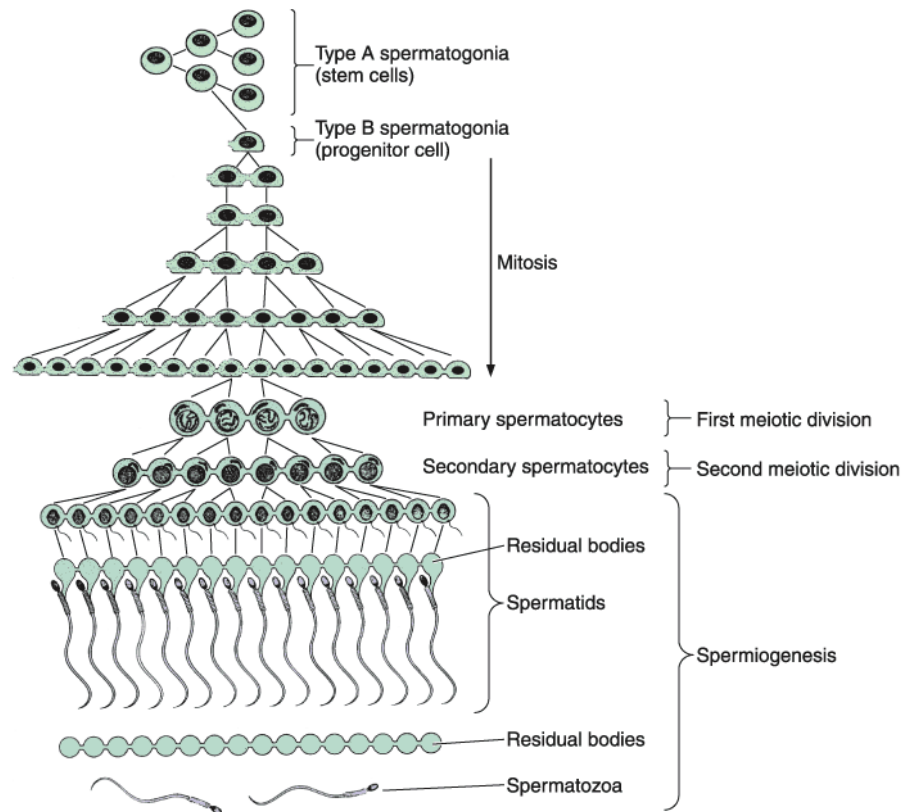
Melatonin is probably not a major regulator of normal sleep patterns, but undoubtedly has some effect. One topic that has garnered a large amount of interest is using melatonin alone, or in combination with phototherapy, to treat sleep disorders. There is some indication that melatonin levels are lower in elderly insomniacs relative to age matched non-insomniacs, and melatonin therapy in such cases appears modestly beneficial in correcting the problem.

Another sleep disorder is seen in shift workers, who often find it difficult to adjust to working at night and sleeping during the day. The utility of melatonin therapy to alleviate this problem is equivocal and appears not to be as effective as phototherapy. Still another condition involving disruption of circadian rhythms is jet lag. In this case, it has repeatedly been demonstrated that taking melatonin close to the target bedtime of the destination can alleviate symptoms; it has the greatest beneficial effect when jet lag is predicted to be worst (e.g. crossing many time zones).

In various species including humans, administration of melatonin has been shown to decrease motor activity, induce fatigue and lower body temperature, particularly at high doses. The effect on body temperature may play a significant role in melatonin's ability to entrain sleep-wake cycles, as in patients with jet lag.

The Biology of Spermatogenesis

Spermatogenesis is the process in which spermatozoa are produced from male primordial germ cells by way of mitosis and meiosis. The initial cells in this pathway are called spermatogonia, which yield primary spermatocytes by mitosis. The primary spermatocyte divides meiotically (Meiosis I) into two secondary spermatocytes; each secondary spermatocyte divides into two spermatids by Meiosis II. These develop into mature spermatozoa, also known as sperm cells. Thus, the primary spermatocyte gives rise to two cells, the secondary spermatocytes, and the two secondary spermatocytes by their subdivision produce four spermatozoa.



Spermatozoa are the mature male gametes in many sexually reproducing organisms. It occurs in the seminiferous tubules of the male testes. Spermatogenesis is highly dependent upon optimal conditions for the process to occur correctly, and is essential for sexual reproduction. DNA methylation and histone modification have been implicated in the regulation of this process. It starts at puberty and usually continues uninterrupted until death, although a slight decrease can be discerned in the quantity of produced sperm with increase in age.

Stages

The entire process of spermatogenesis can be broken up into several distinct stages, each corresponding to a particular type of cell in human. The process of spermatogenesis, after its commencement, goes through a very strictly controlled 3 stages.

1. Mitosis (Spermatocytogenesis)(Proliferative Stage)

In this stage, a diploid spermatogonium, which resides in the basal compartment of the seminiferous tubules, divides mitotically, producing two diploid (Spermatogonium A&B) intermediate cells called primary spermatocytes. Each primary spermatocyte then moves into the adluminal compartment of the seminiferous tubules.

The purpose of this proliferative stage is to ensure the refilling or continuous supply (or a sort of reserve) of germ cells to keep the process of spermatogenesis ongoing. Spermatogonium B, just before it leaves the basal compartment, also undergoes a mitotic process/duplication. This duplication is important for all those recombination processes and genetic assortments which need to occur in Meiosis-I.

2. Meiosis

Primary spermatocytes undergo *meiosis I* to produce two haploid secondary spermatocytes. Which will later divide once more into haploid spermatids. This division implicates sources of genetic variation, such as random inclusion of either parental chromosomes, or chromosomal crossover to increase the genetic variability of the gamete.

Each cell division from a spermatogonium to a spermatid is incomplete; the cells remain connected to one another by bridges of cytoplasm to allow synchronous development. It should also be noted that not all spermatogonia divide to produce spermatocytes; otherwise, the supply of spermatogonia would run out. Instead, certain types of spermatogonia divide mitotically to produce copies of themselves, ensuring a constant supply of spermatogonia to fuel spermatogenesis.

3. Spermiogenesis

When the spermatids are first formed, they still have the usual characteristics of epithelioid cells, but soon they begin to differentiate and elongate into spermatozoa. Each spermatozoon is composed of a *head* and a *tail*. The head, outer covering of the anterior two thirds of the head is a thick cap called the *acrosome* that is formed mainly from the Golgi apparatus. This contains a number of enzymes similar to those found in lysosomes of the typical cell, including hyaluronidase (which can digest proteoglycan filaments of tissues) and powerful *proteolytic enzymes* (which can digest proteins). These enzymes play important roles in allowing the sperm to enter the ovum and fertilize it.

The tail of the sperm, called the *flagellum*, has three major components: (1) a central skeleton constructed of 11 microtubules, collectively called the *axoneme*-the structure of this is similar to that of cilia found on the surfaces of other types of cells 2) a thin cell membrane covering the axoneme; and (3) a collection of mitochondria surrounding the axoneme in the proximal portion of the tail (called the *body of the tail*).

Back-and-forth movement of the tail (flagellar movement) provides motility for the sperm. This movement results from a rhythmical longitudinal sliding motion between the anterior and posterior tubules that make up the axoneme. The energy for this process is supplied in the form of adenosine triphosphate, which is synthesized by the mitochondria in the body of the tail. Normal sperm move in a fluid medium at a velocity of 1 to 4 mm/min. This allows them to move through the female genital tract in quest of the ovum.

♂ Significance

A. **Mitosis**-Continuous supply of stem cells throughout the reproductive life of the male.

B. **Meiosis**

i. Reducing the number of chromosomes in the gametes permits the chromosomal number of zygote to remain

ii. To make sure that the biological door is open to evolution through introduction of controlled variability.

Hormonal Factors That Stimulate Spermatogenesis

1. **Testosterone**, secreted by the Leydig cells located in the interstitium of the testis, is essential for growth and division of the testicular germinal cells, which is the first stage in forming sperm.

2. **Luteinizing hormone**, secreted by the anterior pituitary gland, stimulates the Leydig cells to secrete testosterone.

3. **Follicle-stimulating hormone**, also secreted by the anterior pituitary gland, stimulates the Sertoli cells; without this stimulation, the conversion of the spermatids to sperm (the process of spermiogenesis) will not occur.

4. **Estrogens**, formed from testosterone by the Sertoli cells when they are stimulated by follicle stimulating hormone, are probably also essential for spermiogenesis.

5. **Growth hormone** (as well as most of the other body hormones) is necessary for controlling background metabolic functions of the testes. Growth hormone specifically promotes early division of the spermatogonia themselves; in its absence, as in pituitary dwarfs, spermatogenesis is severely deficient or absent, thus causing infertility.

NB

-**Testosterone** and **FSH** initiate (induce) the process of **spermatogenesis**.

-Testosterone **completes meiosis**.

-**Testosterone is the prime mover** of the process of the process of **spermatogenesis**... (even in the absence of FSH, testosterone can maintain the process of spermatogenesis). This is **very crucial point to remember**.

-But these hormones, no matter how hard they may try, they can facilitate the process but they cannot (shorten) the time duration required for spermatogenesis (that is **74 days**).

The biological effects of androgens

I. Fetal Period/Prenatal/ Prepubertal

- a. Stimulates development of Wolffian duct system.
 - b. **DHT** dependent organogenesis in sex-specific tissues.
(*Masculinize external genitalia*).
 - c. Descent of testes.
 - d. Establishes male pattern of pulsatile release of gonadotropin after puberty.;
- } Male sex organ development

II. Pubertal

- a. Maintenance of secondary sexual characteristics
 - Testes: Sertoli cell maturation, ABP synthesis, spermatogenesis
 - External genitalia: penile and scrotal growth/**DHT**.
 - Growth of body hair: pubic hair (diamond-shaped), facial, chest, back and extremities/**DHT**
 - Nonsexual: eyebrows + extremities

- Ambisexual: axilla
- Sexual: face (moustache + beard), chest + upper pubic triangle

b. Temporal hair recession and baldness/**DHT**

(Baldness: genetic background + $\uparrow$ Androgenic H)

c. Testosteronic voice (bass voice)

- Hypertrophy of the laryngeal mucosa & enlargement of the larynx.
- Thickening of the vocal cords ($\uparrow$ protein synthesis).

d. Bone:

- $\uparrow T \rightarrow \uparrow$ protein synthesis $\rightarrow \uparrow$ bone matrix $\rightarrow \uparrow$ bone growth $\rightarrow \uparrow$ linear growth
- Accelerates epiphyseal fusion of long bones (estrogenic).
- Pelvis: narrow, long, funnel-like shape, strong...($\uparrow$ bone density).

e. Skeletal muscle

- $\uparrow$ Protein synthesis $\rightarrow$ muscular development ($\uparrow$ 50%, shoulder girdle).
- Erythropoiesis
 - Stimulates RBC production
 - $\uparrow$ Epo
 - Act on the erythron in the bone marrow.

g. CNS

- Stimulates libido and other behavioral effects (sexual desire, motivation, social aggressiveness; cognitive function (visuospatial abilities...))

h. Integumentary system

- $\uparrow$ Rate of secretion of sebaceous glands $\rightarrow$ acne, thick (5 α -reductase inhibitor are given to decrease that).

i. Metabolism: $\uparrow$ Basal metabolic rate

j. Fluid + electrolyte balance:

- Retention of N, K, P, Na, H₂O, Ca, SO₄ and PO₄

k. Lipoproteins:

$\uparrow$ VLDL, $\uparrow$ LDL, $\downarrow$ HDL (cardiac/ coronary problems in men). Here Note its difference with female sex hormone

$\downarrow$ Transport proteins: SHBG, CBG and TBG

Seminal analysis (spermatogram)

- Most of its volume is seminal plasma (90%). Sperm cell is only 10%
- Function
 - Nutritional (high fructose)
 - Protective
 - Vaginal acidity (high Ph.)
 - Prevent capacitation
 - Sperm immobility in male tract
- Significance is for screening of infertility
- Sources of seminal plasma
 - Epididymis---5%
 - Bulbourethral--5%
 - Prostate ---30%
 - Seminal vesicles---60%
- Prostate gland
 - PSA (Biomarker of prostatic disease)
 - Citrate (Osmotic balance)
 - Spermine (xic odor)
 - Zn (antibacterial)
 - Ca; increase motility
 - Thromboplastin, fibrinogen, fibrinolysin
 - Cholesterol, phospholipids
 - Proteolytic enzymes
 - Phosphates and bicarbonates act as buffers: Neutralize acid medium of vaginal secretions
- Seminal Vesicles
 - Fructose (1.5-6.5mg/ml, ENERGY source for maintenance of motility).
 - Phosphorylcholine
 - Ascorbic acid/citric acid.
 - Proteinase inhibitor (stabilizes sperm's acrosomal membrane).
 - Prostaglandins (13Pgs).
 - PgE → Makes the cervical mucus more receptive to sperm movt.
 - ↑ Uterine motility which facilitates sperm transport.
 - Semenogelins: induce coagulation of semen after ejaculation.

Hormone Synthesis: Steroid Hormones

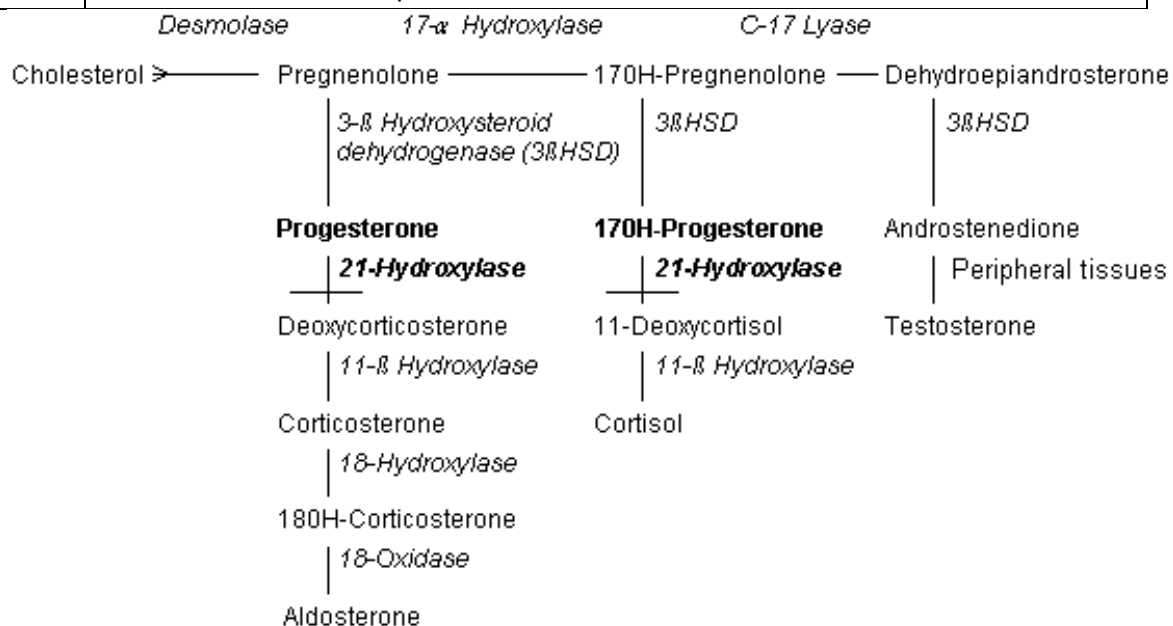
Hormone

- Substance released by an endocrine gland and transported through the blood and act on distance tissue.
- Different type
 - Steroid
 - Peptide
 - Amine

Steroid hormone

- Derived from cholesterol
- Synthesized by specialized organs
 - Adrenal, testes, Ovary

Types of steroids	Androgen Anderstenodion, T, DHEA**,DHEAS* Estrogen,progesterone	Mineralocorticoid	Glucocorticoid
Produced by	Testes→95% T Adrenal cortex Ovary	Zone glomerulosa	Zona fasciculata Zone Reticularis
Production regulation	Adrenal : ACTH Testes and ovary : GTH	K ion concentration Angiotensin To lesser extent ACTH	ACTH
Precursors	Cholesterol ester (de novo 50%) Acetate (plasm pool)		
Key enzyme	<i>17β-hydroxysteroid dehydrogenase*+</i> (androstenedione → testosterone).	<i>21β-hydroxylase + 11β-hydroxylase</i>	<i>21β-hydroxylase + 11β-hydroxylase+++</i>
Clinical correlate	Deficiency of <i>21β-hydroxylase</i> and <i>11β-hydroxylase</i> inhibit the formation of corticoids, with inducing factor(ACTH) presence leading to high production of androgen which in turn lead to virilization and CAH. Deficiency of <i>11β-hydroxylase</i> cause hypertension in addition for there is accumulation of 11-deoxycorticosterone		



- Note
 - ** the major product of adrenal cortex
 - *+ only found in testes; (**androstenedione** → **testosterone**).
 - +++ **does not found in testes so not able to synthesis corticoids**

II. G-protein coupled receptor

- 7-TMS receptor coupled with G-protein
- G-protein
 - 3 sub-units; G- α , G β , G γ dimer
 - 3 types; Gi(G-inhibitory), Gs(stimulatory), Gq

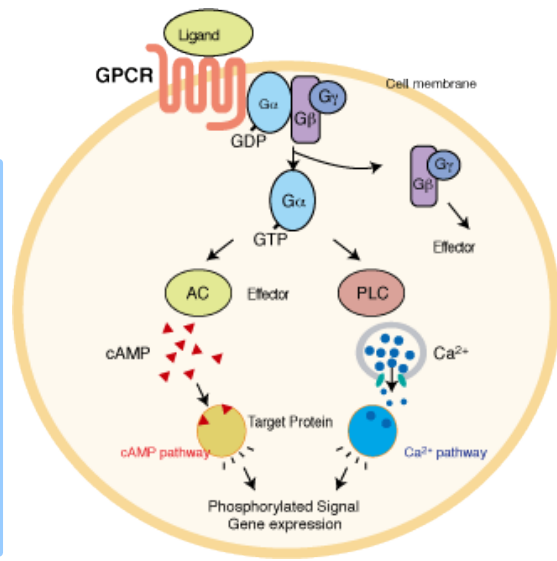
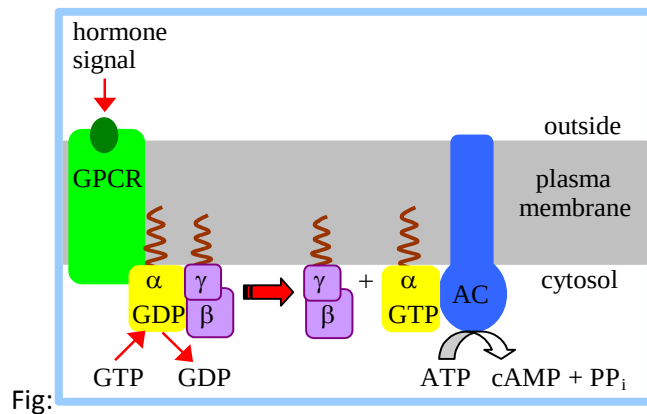
Note

- ** is involved removal of the receptor, promote PDE and cAMP

Clinical correlate

- Cholera toxin
 - Prevents GTP hydrolysis by Gs
- Pertussis toxin
 - Inhibit the exchange GDP for GTP of Gi

Types	G-stimulatory	G-inhibitory	G-q
Enzyme	Stimulate Adenylate cyclase	Inhibitory Adenylate cyclase	Phospholipase
Second messenger	cAMP	-	IP3 and DAG
Effector	Ion-channel	-	IP3: Ca ⁺⁺ SER> DAG:PKC
Hormones acting	epinephrine and glucagon	Ach on heart cell	Vasopressin
Signal shorting	PDE-degrade cAMP α GTPase protein phosphatase β -Arrestin **	Like Gs	Degradation of IP3 Removal Ca by Ca ATPase



Effector Molecules: cAMP, cGMP, Phosphodiesterase and IP₃

Cyclic AMP(CAMP)

- Is the first second messenger discovered
 - Several molecules are involved in the generation, action & degradation of CAMP
1. Adenylyl Cyclase: is an integral cellmembrane enzyme used to produce CAMP from ATP

ATP → CAMP + PP_i

 - Is activated by stimulatory G protien(Gs) & is inhibited by inhibitory G protien(Gi)
 2. Protien kinase A(CAMP dependent protien kinase A)
 - hetrotetramer with 2 regulatory(R & 2 catalytic(C) subunits
 - Is activated by binding of CAMP to its regulatory subunit
 - Each regulatory subunit has a pseudosubstrate sequence(is like the substrate domain of the target prorien but with alanine replacing Ser/Thr)
 - The pseudosubstrate of R binds to & blocks the active site of C subunit
 - Binding of 2 CAMP with the R subunit a connformational change will occur & cause it to dissociate from the C subunit
 - The catalytic subunit can now catalyze the transfer of Pi from ATP to the Ser/Thr of target protiens

3. Phosphodiesterase:

- Once the hormonal stimulus is removed the action of CAMP needs to be terminated & this is done by phosphodiesterase which converts CAMP to 5' AMP
- There are at least 11 types of phosphodiesterase
- It is regulated by its substrates(CAMP, CGMP), hormones & intracellular messengers like Ca^{2+}
- Methylated xanthine derivatives like caffeine inhibit phosphodiesterases & result in increased intracellular CAMP & prolong the actions of hormones

Cyclic GMP(CGMP)

- Is another second messenger
- Made from GTP by the enzyme guanylyl cyclase(has both membrane bound & soluble form)
- It has d/f roles in d/f organs(kidney & intestine – causes change in ion transport & water retention; heart – cardiac mm relaxation, blood vessel – relaxation of smooth muscle)
- Atrial natriuretic peptide(ANP) – released by cells of the atrium when the heart is stretched by increased blood volume
 - Goes to the kidney & blood vessels → activates membrane bound guanylyl cyclase in the collecting duct & vascular smooth mm → CGMP production → increased renal excretion of Na^+ & H_2O & vasodilation
- Endotoxin of E. Coli & other gram negative bacteria can bind to membrane bound guanylyl cyclase in the intestine → CGMP production → increased Cl^- secretion & decreased reabsorption of water → diarrhea
- The cytosolic guanylyl cyclase has a heme group & is activated by nitric oxide(NO)
- In the heart this cytosolic guanylyl cyclase & the CGMP it produces is important for decreasing cardiac contractility by activating Ca^{2+} expelling ion channels
 - Nitroglycerin & other nitrovasodilators used in treatment of Angina work by activating the cytosolic guanylyl cyclase
- The effect of CGMP is terminated by CGMP phosphodiesterase which converts CGMP to 5' GMP
- CGMP phosphodiesterase has different isoforms in different tissues

- Sildenafil(viagra) used for erectile dysfunction inhibits the isoform found in the penis → increase in CGMP → vasodilation of penile blood vessels → erection
- Most of the actions of CGMP are mediated by activation of protein kinase G(CGMP dependent protein kinase)
- It phosphorylates Ser/ Thr residues of target proteins when activated by CGMP(binding of CGMP frees the catalytic subunit from the regulatory subunit).

SEXUALLY TRANSMITTED DISEASES

✱ We prefer the term sexually transmitted infection(STI) than sexually transmitted disease b/c most STIs are asymptomatic.

✱ STIs can be: genital ulcer causing(HSV,syphilis, chancroid), discharge causing (N. Gonorrhea, C.trachomatis, non gonococcal urethritis) & others(HPV- human papilloma virus)

Ulcer Causing:

1. Genital Herpes

Cause: Herpes simplex virus 1 & 2:

- ✱ most cases are due to HSV 2
- ✱ Recurrent & lifelong disease(the virus undergoes latency in our neurons & can't be removed from our body)
- ✱ The primary & recurrent lesions are more severe in HSV 2 than in HSV 1
- ✱ recurrence is more frequent in HSV 2

Mode of transmission - sexual contact with an infected person

- ✱ From mom to newborn via passage through birth canal
- ✱ From mom to fetus(cross placenta & cause still birth or teratogenic effect)
- ✱ Neonatal herpes: In newborns it can cause localized lesions(skin, eye, mouth) or disseminated vesicular lesions, pneumonia, hepatitis, meningitis or encephalitis & has 60-70% mortality rate
- ✱ Risk of transmission is high if mom gets primary infection near term & is low if it is a recurrent infection or if she acquires primary infection early in pregnancy

Herpes Viridae

- ✱ Icosahedral
- ✱ Linear ds DNA viruses

- ✱ Enveloped(hav glycoprotein spikes)
- ✱ Tegment(protein material b/n the envelope & capsid)
- ✱ Are complex(have many proteins coded by their gene)
- ✱ Multiplication & assembly in the nucleus → acquire envelope from the nuclear membrane
- ✱ Although all herpesviridae members have identical morphology(we can not differentiate them by electron microscope) they do not share a common antigen

Herpes Simplex Viruses:

- ✱ Include HSV 1 & 2
- ✱ Virus gets into the body through mucous membrane or broken skin via contact with body fluids or lesions with the virus → virus multiply locally & forms papule on red erythematous base → uniformly sized, fluid filled vesicles → vesicle ruptures & ulcerate → heal in abt 3 wks
- ✱ The fluid filled vesicles contain high amt of the virus + degenerating cells(with intranuclear inclusion bodies) + multinucleated giant cells
- ✱ Lesions may appear in labia, vaginal mucosa, cervix & perineal region(Females) or in prepuce, glans or shaft of penis(Males)
 - ✱ There will be viremia → painful but non suppurative lymphadenopathy
 - ✱ Virus invades sensory nn endings & ascends via their axon & undergoes latency in their ganglia(HSV 1 – trigeminal, HSV 2 – sacral or lumbar) → depression of immunity(stress, cytotoxic drug, menses etc) → reactivation of the virus(multiplies rapidly & descends via the axon to the skin & cause lesions there)
 - ✱ Recurrent Genital Herpes: same manifestations with primary but less severe & heal quickly b/c immunity has already been produced against the virus
 - ✱ Latency helps avoid immune system(no Ab detected in the blood)
 - ✱ Manifestation: can infect many types of cells & cause:
 - ✱ Oropharyngeal lesions, keratoconjunctivitis, encephalitis, genital infection, skin lesions
 - ✱ Apart from the lesions they can also result in systemic manifestations like fever & malaise

Immunity:

- ✱ both cell mediated & Ab mediated immunity are important
- ✱ neonates acquire passive immunity from mom w/c lasts for about 6 months so they are highly susceptible from 6 month – 2 years

Diagnosis:

- ✱ Light microscope(look for intranuclear inclusion bodies & giant cells)
- ✱ Electron microscope might not be that useful b/c all herpes viruses have identical morphology
- ✱ Antigen detection via PCR
- ✱ Cell culture
- ✱ Serological tests for Ab
- ✱ Restriction endonuclease or DNA sequencing can be used to differentiate between HSV 1 & 2 or their serotypes

Treatment:

- ✱ No cure
- ✱ Can decrease the frequency & duration of recurrence
- ✱ Suppression therapy – decreases risk of transmission to partner
- ✱ Drug of choice = Acyclovir
- ✱ Others = famciclovir, idoxuridine, vidarabine
- ✱ nucleotide analogs – inhibit DNA synthesis so inhibit viral replication
- ✱ Vaccines are still under trial

2. **Syphilis:** a highly contagious STI

- ✱ Caused by *Treponema Pallidum pallidum*
 - ✱ *Treponema* has 2 pathogenic species(*T. Pallidum* & *T. Carateum*) & other species which are commensals of the mouth(*T. Orale*, *T. Denticola*, *T. Macrodentium*) & genitalia

***Treponema pallidum*:**

- ✱ Spirochete(thin & spiral shaped with corkscrew movement by axial filaments)
- ✱ Microaerophilic
- ✱ Can't be stained by Gram stain b/c they are very thin
- ✱ We **can not** differentiate the species by morphology, serology, genome analysis
- ✱ Intracellular pathogen
- ✱ Extremely fastidious & require live cell for culture(non pathogenic strain like Reiter's may be cultured artificially on serum enriched broth)
- ✱ Generation time = 33 hrs
- ✱ Virulence factor- hyaluronidase, fibrinogen & outer membrane proteins

- * Easily inactivated by drying, heating (fever therapy for syphilis), oxygen, disinfectants, bismuth, mercurials, soap, antibiotics
- * Strictly human pathogen

Mode of transmission:

- * Sexual contact (major form)
- * Blood transfusion (can be prevented by storing the blood for at least 4 days @ 0-4°C) – no primary chancre
- * Vertically from mom to baby
- * Organ transplantation
- * Indirect (hand contact of health care workers with patients' lesion) – extragenital chancre

Pathogenesis: mainly due to the host's immune response

- * It is primarily a disease of blood vessels & perivascular areas but it can invade almost any organ
- * T. Pallidum enters via mucous membrane or abraded skin & binds to the epithelium → multiplies locally → penetrates into the blood & lymphatics → multiplies & spreads through the body

Signs & Symptoms: “great imitator” – b/c has manifestations similar to many other diseases

1. **Acquired Syphilis:**

- * The infectious dose is as low as 60 & if untreated the disease will progress through 3 stages
- A. Primary syphilis:
 - * Hard painless chancre (due to treponema multiplying at the site of inoculation) – mainly on the genitalia, mouth or nipples
 - * Chancre – hard, painless, indurated, avascular & superficially ulcerated papule & is covered by a thick exudate which is rich in bacteria
 - * regional lymphadenopathy (swollen, discrete but non tender lymph nodes)
 - * incubation period (IP) – about 3 weeks
 - * chancre heals spontaneously within 3-6 weeks but may persist in HIV patients

B. Secondary Syphilis – due to wide spread distribution of the virus via blood

- * Skin rash (maculopapular or roseolar) + condyloma lata (due to the spread of treponema to the skin & mucous membrane)
- * The rash involves the whole body including the palms & soles

- ✱ Condyloma lata – are thickened, flat & raised lesions seen usually around the anus, vulva & in mucous patches of the mouth
- ✱ Systemic symptoms: fever, malaise, headache, arthralgia, weight loss, sore throat
- ✱ Alopecia might be seen in some patients
- ✱ It may result in acute meningitis, hepatitis, nephropathy, arthritis, periostitis, GIT & eye manifestations
- ✱ NB: treponema is found in the lesions and in the blood in high amount in primary & secondary syphilis as a result this stages are highly contagious

C. Latent Syphilis- no clinical symptoms but infection is still in the body (so patient is still infective) & we can detect antibodies in the blood

- ✱ Some cases (25%) can show relapse to secondary syphilis; only 1/3 of untreated cases will progress to tertiary syphilis

D. Tertiary Syphilis: non contagious but highly destructive phase

- Develops over many years

- Has 3 forms;

- ✱ Gummatous Syphilis: gummas are localized granulomatous lesions which mostly occur in the skin & bone with a necrotic center covered by fibrotic capsule
- ✱ Neurosyphilis: degenerative changes like paresis, tabes dorsalis (unsteady gait, lightning pain & urinary incontinence). There could be late neurosyphilis with seizure & dementia
- ✱ CVS syphilis: aneurysms of the ascending aorta & aortic arch + aortitis + aortic valve insufficiency
- ✱ NB: antimicrobial treatment can only reverse the gummatous lesions but not the cardio & neurosyphilis

2. **Congenital Syphilis:** due to infection of the fetus via placenta from mom

- ✱ Has high mortality rate & even if the baby survives it will develop early or late congenital syphilis
 - A. Early CS- within 2 years after birth
 - ✱ Pemphigus syphiliticus (vesiculobullous eruption), condyloma lata, hepatosplenomegaly & jaundice, lymphadenopathy, otitis, desquamation of palm & plantar, snuffles (Nasal discharge which is highly infectious), Irritability and Difficulty feeding.
 - B. Late onset: after 2 years
 - ✱ Hutchinson's triad (63%): Hutchinson teeth (centrally notched & widely spaced upper incisors), interstitial keratitis (blindness, 5-20 years later) and VIII nerve (deafness, 10-40 years). Additionally, Frontal bossae (bony prominence of forehead), bone deformities like Saddle nose,

Perforation of palate, Saber shins, Clutton's joint (bilateral painless swelling of the knees), Short maxilla, Protruding mandible and High palatal arch.

- ✱ CS does not damage the fetus until 4 months of pregnancy so treating the mom before 4 months of gestation can prevent CS

Diagnosis:

- ✱ Direct examination of treponema from samples taken from chancre, rash or condyloma lata via darkfield microscope (we see helical shaped treponemes with corkscrew movement) or via fluorescent microscope (by using fluorescein labelled anti treponemal Abs)
- ✱ Serological tests:
 1. non treponemal or Reagin test
 - ✱ We mix patient's serum with cardiolipin Ag (from beef heart) & try to detect if there is anticardiolipin Abs (reagin = Ig M or Ig A) in the serum
 - ✱ Non specific - b/c other diseases like viral hepatitis, malaria, infectious mononucleosis can cause production of anticardiolipin Abs
 - ✱ 2 techniques; Complement fixation (Wasserman reaction) or Flocculation test (VDRL or RPR)
 - ✱ Used for screening (b/c of low cost) & monitoring treatment efficacy (b/c it returns to normal 6-18 months after treatment)
 2. Treponemal antigen tests: antigens for the test are prepared from virulent treponemes
 - ✱ Are specific b/c we are looking for antibodies specifically produced for treponema
 - ✱ More complex & expensive
 - ✱ Confirmatory test
 - ✱ Include Indirect fluorescent treponemal absorption antibody test (FTA Abs test), treponema pallidum hemagglutination test (TPHA) & treponema pallidum immobilization test (TPI)

Treatment:

Drug of choice – Penicillin (for both acquired & congenital)

- ✱ Tetracycline or erythromycin can be substituted.
- ✱ Jarisch Herxheimer reaction: occurs within 2 hours post treatment due to release of endogenous pyrogens from dying spirochetes
- ✱ Includes fever, chill, malaise, headache, muscle aches

3) Chancroid:

- ✳ Caused by *Haemophilus ducreyi*
- ✳ Small, gram negative coccobacilli
- ✳ Fastidious (requires X factor or hematin for growth) & labile

Signs & symptoms:

- ✳ Soft chancres (painful & may be single or multiple)
- ✳ Large & tender lymph nodes which might be pus filled & even rupture
- ✳ First a tender papule with an erythematous base develops → ulcerates and may join other lesions & enlarge
- ✳ There will be intense inflammatory response in the lymph nodes

Diagnosis – usually clinical

Treatment:

- ✳ Responds well to erythromycin, azithromycin, ceftriaxone but there could be R plasmid mediated drug resistance

Discharge Causing:

1) **Gonorrhea;**

- ✳ Caused by *Neisseria gonorrhea*
- ✳ Gram negative kidney shaped diplococci
- ✳ Non motile, fastidious (grows on chocolate agar or modified Thayer-Martin medium which is a chocolate agar with antibiotics)
- ✳ Most are catalase positive & all are oxidase positive
- ✳ Can be differentiated from *N. meningitidis* by their CHO fermentation property (both can ferment glucose but only *N. meningitidis* can ferment maltose)
- ✳ Obligate human pathogens; most are found intracellularly but may also be found extracellularly
- ✳ Susceptible to cold & drying
- ✳ Most have R plasmids → resistant to penicillin & tetracyclins
- ✳ Virulence Factors: pili (adherence & antiphagocytic), IgA protease, Opa proteins (bind to CD4+T cells & impair immunity), lipooligosaccharide (LOS), transferrin/lactoferrin binding protein (captures host's iron)

Pathogenesis:

- * Binds to nonciliated epithelial cells(especially prefers urethra, cervix, pharynx, conjunctiva) by its pili → can penetrate to the submucosal layers & bring strong influx of PMN

Manifestation:

- * Although asymptomatic infections can occur in both sexes it is more common in women
- * Men = urethritis(inflammation of urethra, purulent urethral discharge, dysuria)
- * Some men with gonorrhea are asymptomatic & are more likely to develop complications like epididymitis(unilateral testicular swelling & pain), urethral stricture, prostatic abscess, orchitis
- * Women = is asymptomatic 1/3 of the time; usually results in cervicitis(inflammation of cervix, purulent vaginal discharge, abnormal bleeding) but can also spread to the urethra(urethritis)
- * Complications in women include accessory gland infections(bartholinitis), PID(due to upward spread of the gonococci most probably by riding on sperm) & Fitz-Hugh-Curtis syndrome(perihepatitis due to direct inoculation of the gonococci on the surface of the liver which causes adhesion between the liver & the peritoneum)
- * Scar tissue formation as a result of PID may result in infertility or ectopic pregnancy
- * Discharge of gonorrhea is thick, yellow green & smelly

Disseminated Gonococcal Infection(Gonococcemia) – is an infrequent disseminated systemic infection with septicemia, fever, migratory arthralgia, purulent arthritis, pustular rash over an erythematous base (on the extremities only). It may also damage heart valves

- * DGI is more common in women b/c asymptomatic gonorrhea which is left untreated is more common in women

Ophthalmia Neonatorum: a destructive eye lesion(erythematous, edematous eyelid with purulent discharge) of the new born acquired as the baby passes through infected mom's birth canal

Diagnosis:

We can take sample(discharge) & do gram stain(gram negative diplococci within PMNs), culture(on chocolate agar or modified Thayer Martin or modified NYC medium) & nucleic acid based tests(to differentiate between N. Gonorrhea & C. Trachomatis)

Treatment:

- * Drug of choice is cephalosporins(ceftriaxone, cefixime) b/c of resistance to penicillins, tetracyclins, sulfonamide + rising resistance to fluoroquinolones too
- * Treatment regimens usually include drugs against C. Trachomatis(azithromycin or doxycycline) too b/c the 2 are frequently found together

- ✱ No permanent immunity following infection

2) **Non Gonococcal Urethritis(NGU):**

- ✱ The most common cause of urethritis among heterosexual men
- ✱ It often occurs together with gonococcal urethritis
- ✱ Causes: C. Trachomatis → Mycoplasma genitalum → trichomonas vaginalis(decreasing order)
- ✱ NGU is mostly asymptomatic but if symptoms occur they include mucoid to clear urethral discharge, dysuria, cervicitis(women)

A) Chlamydia Trachomatis:

- ✱ Thin, Gram negative, obligate intracellular bacteria
- ✱ Major cause of NGU(serovar D-K)
- Complications;
- ✱ Men – epididymitis, Reiter's syndrome(arthritis + conjunctivitis + urethritis)
- ✱ Women - PID, Fitz-Hugh-Curtis Syndrome, rarely Reiter's syndrome
- ✱ Can cause ophthalmia neonatorum in neonates

Diagnosis: it is difficult to do gram stain

- ✱ culture(needs cell culture)
- ✱ Non culture tests(PCR, complement fixation, rarely serology)

Treatment: azithromycin, doxycycline, erythromycin

Lymphogranuloma Venerum: a chronic disease of the lymphatic system caused by C. Trachomatis serovar L1, 2 & 3

- ✱ Transmitted sexually
- ✱ More common in men
- ✱ Self limited, painless papule or ulcer → painful inguinal & femoral lymphadenopathy
- ✱ Has 3 stages:

Primary stage: painless papule or ulcer which heals spontaneously, does not leave any scar but has widespread dissemination

Secondary stage: painful regional suppurative lymphadenopathy(buboes) that may drain pus + other symptoms(conjunctivitis, fever, chill, arthritis, CNS symptoms)

Tertiary stage: (urethro-genital perineal syndrome) - extremely painful inguinal lymphadenopathy (LAD), lymphatic obstruction & non-destructive elephantiasis of the genitals, rectal stenosis, fibrosis & scarring of genital tract. Women may develop fistulas b/n vagina & rectum.

Complications: urethral stricture, reiters syndrome, disfiguring enlargement of the external genitalia

Diagnosis:

- ✱ Lymph node biopsy(granulomatous inflammation & necrosis)
- ✱ Nueclic acid tests
- ✱ Blood tests

Treatment: doxycycline, erythromycin

B) Mycoplasma & Ureaplasma:

- ✱ There are 5 species which cause human disease in the family mycoplasmatacie
- ✱ 3 are in the genus mycoplasma(Mycoplasma hominis, M. Genitalum & M. pneumoniae) while 2 are in the genus ureaplasma(U. Urealyticum & U. parvum)
- ✱ They lack cell wall
- ✱ Mycoplasma genitalium: an emerging sexually transmitted pathogen which can cause urethritis in men and cervicitis, PID & Infertility in women. M. genitalium is responsible for about 20%–35% of non-chlamydial NGU cases.
- ✱ Mycoplasma hominis: causes respiratory infections and Genital infections such as pyelonephritis(in males)and female (PID, Infertility, postpartum fever and fever following abortion as complication of endometritis).

Diagnosis: we use PCR since it is difficult to culure it

Treatment: tetracycline & erythromycin

Ureaplasma:

- ✱ Men - causes NGU(40% of NGU) & UTI
- ✱ Women- PID & UTI
- ✱ Person to person transmission
- ✱ Although it can occur at any age highest rates are seen at age 5-20
- ✱ High attack rates are also seen in closed populations (military recruit barrack, boarding schools, etc).

Clinical manifestations:

- ✱ Incubation period: 2-3 weeks
- ✱ Most individuals are asymptomatic, some show urethritis

Diagnosis - Organisms may be cultured for weeks-months

- ✳ Detection of Ags or specific nucleotide sequence
- ✳ M. hominis & U. urealyticum can be detected in culture within 2-5 days.
- ✳ Urease test: used to differentiate Mycoplasma from Ureaplasma species (Mycoplasma are urease negative and Ureaplasma are urease positive).

Treatment

Mycoplasma are sensitive to: Tetracycline, Erythromycin, Azithromycin and Ofloxacin. They are resistant to beta lactam antibiotics like penicillin & cephalosporins, because mycoplasma lack cell walls

C) Gardnerella Vaginalis & other anaerobic flora:

Gardnerella vaginalis (former names- haemophilus or corynebacterium vaginalis)

- ✳ Gram negative to gram variable rod (pleomorphic), non motile
- ✳ Facultative anaerob that grows best when supplemented with CO₂
- ✳ Produces a cytotoxin called vaginolysin (VLY) & Causes bacterial vaginosis (BV)

Bacterial vaginosis:

- ✳ is a polymicrobial vaginal infection where there is reduction in lactobacilli (normal vaginal flora) & an increase in Gardnerella & other anaerobes like Mobiluncus, Bacterioides, Peptostreptococcus
- ✳ Is called vaginosis b/c it does not have true vaginal inflammation
- ✳ Not considered as a classical STD b/c it can occur spontaneously

Pathogenesis

- ✳ Normal vagina has lactobacilli. It produces lactic acid & H₂O₂ which make the vagina acidic & prevents the growth of other microorganisms. Loss of protective lactobacilli may lead to BV by allowing other bacteria like Gardnerella to overgrow.
- ✳ Risk Factors: Multiple or new sexual partners, Smoking, Genetic predisposition. broad-spectrum antibiotics (may destroy normal flora), Douching, Overused/retained tampons, Wearing IUDs (contraceptive sponges disrupt balance) and decreased estrogen production (Ex. menopause) b/c estrogen stimulates glycogen uptake by vaginal epithelial cells & glycogen serves as the main food source for lactobacilli.

- ✱ BV is more common in pregnant women than in non-pregnant population (15-30%). & can cause premature labor/delivery and post-partum endometriosis. There is also a controversial association with miscarriage.

Complications: increased risk for PID, Ectopic pregnancy, premature birth, Low birth weight in infants born to infected mothers & increased risk for other STD including: Chlamydia trachomatis, Gonorrhea, Herpes simplex virus (HSV) and HIV infections.

Signs & Symptoms:

- ✱ G. Vaginalis is isolated from 92-98% of BV.
- ✱ Most cases of BV (50-75%) are asymptomatic or mild(thin, grayish-white & fishy vaginal odor, itching, irritation)

Diagnosis: mostly underdiagnosed b/c most women think they have a yeast infection

- ✱ Amsels Criteria (We take a sample of vaginal discharge & test for the following):
 - ✱ pH > 4.5
 - ✱ Presence of > 20% clue cells on wet mount examination(under high power field)
 - ✱ Positive Whiff test(amine odor when discharge is mixed with 10% KOH)
 - ✱ Homogenous, nonviscous, milky white discharge
- NB** if 3 out of 4 criterias are fulfilled then it is positive for BV
- ✱ Nugent Score test: We can also count the different types of bacteria in the discharge & grade it
- ✱ We can also culture the bacteria

Treatment:

- ✱ Metronidazole, clindamycin, amoxycillin
- ✱ No permanent immunity(recurrence can occur)

3. Vulvovaginal Candidiasis

- ✱ 2nd most common cause of vaginal symptoms after BV
- ✱ Caused by Candida albicans
 - Unicellular yeast, can form pseudohyphae & hyphae(but is not dimorphic b/c both yeast & hyphae can be found in tissue)
 - Multiplies asexually by budding

- Virulence factors: surface molecules, lytic enzymes(Aspartyl protease), undemanding nutrient requirement & fast growth rate, surface hydrophobicity, molecular mimicry etc
- * Normal flora in about 35% of women & usually occurs as a result of disruption of the normal vaginal flora due to intensive antimicrobial treatment
- * Use of oral contraceptives, late pregnancy & poorly controlled glucose in DM can be risk factors
- * It involves both the vagina & vulva
- * Most cases are autoinfections; is not transmitted sexually & it does not spread from person to person except in ICU where crossinfections may occur

Signs & symptoms:

- * Intense & unremitting itching & burning, scanty, whitish & curdy vaginal discharge, the area involved may become red & swollen

Diagnosis: specimen = vaginal discharge

- Microscopy, gram stain, culture, germ tube test

Treatment

- * Intra-vaginal nystatin or clotrimazole or oral fluconazole
- * Prevention: minimizing the use and duration of antibacterial medications, effective treatment of underlying conditions such as diabetes

4. Trichomoniasis:

- * Caused by *Trichomonas vaginalis*
 - Flagellated protozoa
 - Has only trophozoite stage
 - Divides by longitudinal binary fission
 - Human parasite only
 - Inhabits vagina(F) & urethra & prostate(M)
- * 3rd most common cause of vaginal symptoms

Transmission: mainly sexually, mom to child during birth, non sexual contact may possibly transmit it too

Pathogenesis:

- * damages epithelial cells by direct contact & release of cytotoxic substances
- * binds to host plasma proteins & evade recognition by the immune system
- * phagocytoses vaginal cells & vaginal bacteria(vaginal pH increases as a result of infection)

Syptoms:

- ✳ Infected men are usually asymptomatic; signs & symptoms are usually seen in women
- ✳ Itching of vulva & vagina, frothy, malodorous, yellow green vaginal discharge + sometimes dysuria
- ✳ If men are symptomatic they develop dysuria, penile discharge, painful testes & inflammation of prostate
- ✳ Trichomoniasis during pregnancy may be associated with preterm labour, low weight infants, respiratory & genital infection of the newborn. It can also increase vertical transmission of HIV due to disruption of the vaginal mucosa
- ✳ People with trichomoniasis are 2x more likely to get infected with HIV
- ✳ Untreated trichomoniasis may result in increased risk for cervical CA & PID in females & infertility in males(due to decreased sperm motility & viability)

Diagnosis: physical examination, wet mount, culture, giemsa, PAP smear, antigen & antibody tests

Treatment: Metronidazole; alternatives- tinidazole, clotrimazole

- ✳ Sexual partners should also be treated + no sex during treatment

Other STDs:

Granuloma inguinale

- ✳ uncommon STD characterized by hard, granulomatous proliferating, deep, purulent ulcers on or near the genital organs
- ✳ begins as a small sore but may spread to involve large areas as ulcerating form.
- ✳ Is also called Donovanosis b/c it forms Donovan bodies(intacytoplasmic inclusion bodies in macrophages)
- ✳ Caused by *Klebsiella granulomatis*, formerly called *Calymmatobacterium granulomatis*
- ✳ Enterobacteriaceae family,
- ✳ Gram-negative, pleomorphic coccobacilli, facultative anaerobe & Non-motile
- ✳ Intracellular(found in cystic spaces in large mononuclear cells)
- ✳ Safety pin appearance(b/c of condensation of chromatin)

Transmission

- ✳ Mainly via sexual contact; but repeated exposure is necessary

- ✳ Fecal route or by passage through an infected birth canal

Signs & symptoms:

- ✳ Incubation period: 1 week-3 months
- ✳ Ulcers progressively expand- locally destructive
- ✳ The regional lymph nodes not involved (pseudobubo), no LAD (lymphadenopathy).
- ✳ 4 main types of cutaneous lesions: Nodular, Ulcerovegetative(most common), Cicatricial, Hypertrophic or verrucous

Complications

- ✳ Secondary infections, rectovaginal fistula, urethral strictures, cystitis, genital edema, SCC & BCC(in long-standing lesions/scars)

Diagnosis

- ✳ Extremely fastidious(Grown on egg yolk, and Modified Levanthal agar)
- ✳ Donovan bodies with bipolar staining (safety-pin appearance) from smear from ulcers
- ✳ Wright's or Giemsa stained smears(safety pin appearing rods with pink capsule)
- ✳ PCR
- ✳ Papanicolaou smears may show Donovan bodies

Treatment: Trimethoprim/sulfamethoxazole or doxycycline. Alternatives: ciprofloxacin, erythromycin, azithromycin. The antibiotic should be given for at least 3 weeks.

Human Papilloma Virus: family = papilloma viridae

- ✳ DNA virus(ds & circular), icosahedral & non enveloped
- ✳ Has oncogenic potential(most cases of cervical cancer + can also cause penile & anal cancer)
- ✳ Difficult to study(b/c no tissue culture)
- ✳ About 130 types of HPV have been identified based on molecular characteristics
- ✳ Love epithelial cells of the skin & mucous membrane
- ✳ Can cause different types of warts(skin, plantar, flat wart, genital wart or condyloma acuminata, laryngeal papilloma)
- ✳ HPV 6 & 11 are low risk groups associated with benign lesions while HPV 16 & 18(high risk groups) mainly cause cervical cancer

Pathogenesis: HPV 6 & 11 infect & replicate in keratinocytes of skin & mucous membrane → it makes the keratinocytes multiply rapidly & results in formation of genital warts

- ✱ koilocytic changes is seen in the keratinocytes(enlarged keratinocytes with a halo around a smaller nuclei)
- ✱ genital warts remain localized & mostly recede spontaneously; recurrence is common
- ✱ HPV 16 & 18 also infect keratinocytes & then integrate into the hosts chromosome → expression of E6 & 7 genes and their translation into proteins that damage P53 & Rb protein(tumour suppressors) → keratinocytes multiply rapidly & are now more susceptible to mutations → cancer

Diagnosis:

- ✱ microscopic(histologic appearance of warts), PCR, PAPs smear(detect koilocytic changes in cervical epithelium), rarely Ab test

Prevention:

- ✱ quadrivalent vaccine(HPV16,18,6 & 11) available for girls 9-26 yrs(100% effective if given before sexual exposure but useless in those already exposed to the virus) 3 shots over a period of 6 m

Hepatitis:

- ✱ Is an inflammation of the liver
- ✱ Has many causes but the most common one is viral hepatitis
- ✱ 5 RNA = Hep A,C,D,E,G
- ✱ 1 DNA virus = Hep B
- ✱ Hep A & E(the only non enveloped ones) are transmitted feco-orally while the rest are parenterally(blood)
- ✱ Viral hepatitis can be of 2 types: acute & chronic
- ✱ Acute: sudden onset, mild-severe course, complete resolution
- ✱ Chronic – prolonged course of active disease or silent asymptomatic disease
- ✱ Acute hepatitis can be caused by all the above viruses while chronic is caused only by Hep B,C & D
- ✱ Hep C & E are the only ones with no vaccine

Hepatitis B virus: family= hepadnaviridae

- ✱ Partially double stranded DNA virus(positive strand is not full length)
- ✱ Enveloped, icosahedral

- ✧ The only DNA virus with reverse transcriptase
- ✧ The intact virus is called Dane particle(spherical structure with an envelope + icosahedral capsid + core with double stranded DNA & DNA polymerase enzyme)
- ✧ Hepatitis B surface Ag(HBsAg) – spherical to filamentous particles composed of the envelope & some capsid proteins which have dissociated from the virion
- ✧ Does not contain the DNA so is not infectious
- ✧ Antibodies against HbsAg are protective
- ✧ Hepatitis B core Ag(HBcAg) – made of the viral core
- ✧ Is also antigenic but antibodies against it are not protective
- ✧ HbeAg – a soluble component of the core Ag released only during active infection where there is high viral replication & high infectivity
- ✧ Since it is partially double stranded the initial replication is to complete the circular DNA which is followed by transcription to mRNAs & translation to viral proteins
- ✧ One of the mRNAs is reverse transcribed into a DNA which will be the genetic material of the progeny
- ✧ Some DNA will integrate into the host's DNA & causing a carrier state

Mode of transmission:

- ✧ Mainly by contact with blood. Although they are not as infectious as blood, other fluids like semen, vaginal discharge, saliva, sweat etc may also transmit the disease
- ✧ Sexual contact with acute, carrier states
- ✧ Perinatal

Signs & Symptoms:

- ✧ IP = 2-5 months
- ✧ 50-60% of adults with HepB infection are asymptomatic
- ✧ If symptomatic – jaundice, fatigue, nausea, vomiting, abdominal pain, loss of appetite, joint pain

Diagnosis:

- ✧ Antigen detection(ELISA)
- ✧ Antibody detection(ELISA)
- ✧ DNA detection(PCR)

Treatment:

- ✧ Interferon alpha, Lamivudine, Adefovir dipivoxil

Prevention:

- ✧ Safe sexual practice, screening blood before transfusion, vaccination
- ✧ Hep B vaccine is a recombinant DNA vaccine which confers premanent immunity
- ✧ 3 injections(the 2nd injection is given 1 month after the 1st & the 3rd is given 6 months after the first)

Hepatitis C virus: family = flaviviridae

- ✧ Single strands RNA (+ sense)
- ✧ Icosahedral, enveloped
- ✧ Has highly variable envelope glycoprotien(6 serotypes & > 100 subtypes) = made developing vaccine difficult

Transmission:

- ✧ parentral(via contact to blood), perinatal, unsafe sex(low efficiency)

Signs & syptoms;

- ✧ IP= 6-8 wks
- ✧ The acute form is milder when compared to Hep B(about 80% remain asymptomatic & even if syptoms develop they are mild - jaundice, nausea, abdominal pain, loss of appetite, dark urine)
- ✧ About 50% of the patients develop chronic hepatitis & a large percentage of this develop chronic active hepatitis & cirrhosis
- ✧ There are certain risk factors that promote the progression and severity of chronic hepatitis C :
 - ✧ Alcohol, age > 40 @ time of infection, HIV coinfection, chronic HBV coinfection, male

Diagnosis:

- ✧ ELISA
- ✧ Confirmatory testing: **PCR, RIBA**(recombinant immunoblot assay)
- ✧ Genotype testing done when treatment anticipated

Treatment: Interferon, Ribavirin(response rate = 40-50%); Vaccine is still under trial

HIV & AIDS:

- ✧ Caused by the viruses HIV 1 & HIV 2 which are retroviruses in the subfamily of lentiviruses(nononcogenic retroviruses)

- ✱ Most cases are due to HIV 1 while HIV 2 is mainly localized in W. Africa & is less virulent(the 2 differ by genome organization & evolutionary relation with other lentiviruses)
- ✱ Based on ENV gene sequencing there are 3 groups of HIV1(M,N,O). The predominant group(M) has 10 subtypes or Clades(A-J); Sub type C is the most common one in Ethiopia & India
- ✱ HIV 2 has 5 subtypes(A-E)

HIV virus is:

- Enveloped & spherical with single-stranded RNA(linear, diploid & + sense)
- Helical nucleoprotein within icosahedral capsid & reverse transcriptase enzyme
- 3 genes GAG – POL – ENV, Three polyproteins, Two glycoproteins (gp160): gp120 and gp41 & may carry oncogene
- Rapidly inactivated by extreme pHs, alcohol, bleach, lysol, H2O2 etc
- ✱ Untreated HIV involves 3 stages of the disease & in many patients symptoms of HIV infection do not occur until stage 3(8-10 yrs after infection & death occurs about 2 yrs later)
- ✱ Stage 1: has an IP of 1-3 weeks; symptoms may not occur but if they do they include mononucleosis like symptoms(malaise, pharyngitis, lowgrade fever, cervical lymphadenopathy), myalgia, arthralgia, lymphadenopathy, hepatosplenomegaly etc; this stage ends with the production of high titers of Ab at 2-3 months postinfection
- ✱ Stage 2: is usually asymptomatic & persists mostly for 6 yrs or longer; patient produces large amount of anti-HIV antibody; HIV is detectable in blood, semen & cervical secretions; if symptoms occur they involve lymphadenopathy or AIDS related complex(ARC) – fever, fatigue, chronic diarrhea, weight loss, night sweat, immunologic abnormalities,oral candidiasis, dementia & spontaneous neoplasms
- ✱ Stage 3: where symptoms of d/f opportunistic infections due to bacteria, virus, fungi, protozoa & spontaneous neoplasms occur(the severity of this stage is related to the decline in CD4+ T cells)
- ✱ The gp120 binds to CD4 and a chemokine receptor (CCR5 or CXCR4). Binding of gp120 to its two receptors brings the virus close enough to allow gp41 to bind to the host cell membrane and cause fusion of HIV with the host cell membrane
- ✱ Infected T4 cells express high levels of HIV envelope glycoprotein on their surface, this lead to fusion with neighboring uninfected T4 cells.
- ✱ This is followed by lysis of large numbers of fused cells & rapid depletion of T4 helper cells
- ✱ Loss of immune function occurs as a result of loss of CD4+ T cells; in advanced stages loss of cytotoxic CD8+ T cells also occurs

- ✱ The virus can undergo latency in lymphocytes & monocytes. Activation may be due to coinfection with other viruses(HSV, HBV, EBV, CMV)
- ✱ **Pediatric AIDS:** has very poor prognosis b/c immune system has not developed well; they develop signs & symptoms 2 years postinfection & die about 2 years later
- ✱ HIV can be transmitted by unsafe sex, blood transfusion & from mom to baby: High titers of HIV are found in two body fluids—blood and semen.
- ✱ Transmission of HIV is thought to be cell mediated & not the result of free virus in a cellular fluids.

Diagnosis:

- ✱ Viral isolation on tissue culture
- ✱ Antibody detection by ELISA(if positive it must be confirmed by western blot technique)
- ✱ Antigen or nucleic acid detection by PCR
- ✱ Pediatric HIV: plasma HIV-1 RNA test

Treatment:

- ✱ To avoid development of resistant HIV variants, a combination of at least three drugs from at least two classes is usually administered. Although there are d/f combinations the ones associated with better prognosis are: 2 nucleoside analog reverse transcriptase inhibitors(NRTI) + 1 non nucleoside analog RT inhibitor or 2 NRTI + 1 protease inhibitor
- ✱ Successful HAART can suppress viral replication & halt damage to the immune system but it needs to be taken regularly & in combination to avoid development of resistance
- ✱ No vaccine till now b/c of high mutation of virus

TORCH

- ✱ is an acronym for a set of perinatal infections that can lead to fetal anomalies & even death
- ✱ They gain access transplacentally via the chorionic villi (transmission may occur at anytime during gestation or occasionally at the time of delivery via maternal-to-fetal transfusion)
- ✱ T: TOXOPLASMOSIS
- ✱ O: Other infections (Syphilis, Varicella Zoster, Parvovirus B- 19, HBV)
- ✱ R: RUBELLA
- ✱ C: CYTOMEGALOVIRUS
- ✱ H: HERPES SIMPLEX VIRUS - 2

Toxoplasmosis

- ✳ Caused by an intracellular parasite, *Toxoplasma Gondii*.
- ✳ Human acquire the infection by: oocyst contaminated soil, salads & vegetables, raw or undercooked meat containing tissue cysts (Sheep, pigs and rabbits are the most common meat sources), may also be related to consumption of unfiltered water.
- ✳ Pregnant mom infected for the 1st time can transmit it to the fetus; Infection rate higher with infection in 3rd trimester but the outcome is severe in 1st trimester

Signs and symptoms

- ✳ Infected pregnant women are usually asymptomatic. Some may have a mild mononucleosis-like syndrome, regional lymphadenopathy, or occasionally chorioretinitis.
- ✳ Manifestations in the infant infected after the first trimester include: epilepsy, encephalitis, microcephaly, intracranial calcifications, hydrocephalus, psychomotor or mental retardation, chorioretinitis, blindness, anemia, jaundice, rash, pneumonia, diarrhea and hypothermia.
- ✳ Infants may be asymptomatic at birth & develop the disease months to years later. Most often these children develop: chorioretinitis with or without blindness or other neurologic problems, including retardation, seizures, microcephaly and hearing loss.
- ✳ The classic triad of findings: chorioretinitis, hydrocephalus and intracranial calcifications

Diagnosis

- ✳ Serial IgG measurement (for maternal infection)
- ✳ Amniotic fluid PCR (for fetal infection)
- ✳ Serologic testing, brain imaging, CSF analysis and ophthalmologic evaluation (for neonatal infection)
- ✳ PCR testing of various body fluids or tissues

Treatment

- ✳ In pregnant women with an established recent infection, SPIRAMYCIN (3g daily in divided doses)
- ✳ In neonates: Pyrimethamine, + Sulfadiazine + Folinic Acid

PARVOVIRUSES

- ✳ Smallest of all human viruses ; with several genus
- ✳ Human pathogen B19, no other host
- ✳ linear, single stranded DNA

- ✳ Non-enveloped, icosahedral capsid
- ✳ Nuclear replication, depending on dividing host cell (S-phase)
- ✳ Transmission: Respiratory, blood transfusion, urine, feces, vertical (mother- fetus)

Pathogenesis & Pathology

- ✳ The principal targets for human B19 parvovirus are immature cells in the erythroid lineage
- ✳ major sites of replication = the adult marrow, some blood cells, and the fetal liver.
- ✳ Viral replication → cell death → decreased red cell production.
 - In immunocompromised patients, persistent B19 infections occur, resulting in chronic anemia.
 - In cases of fetal liver death, chronic infections may have caused severe anemia in the fetus.
 - IgM or IgG appear in the blood and Immune complex also contribute for pathogenesis.

Infection during Pregnancy

- ✳ 5% of maternal infections are in the first half of the pregnancy. Maternal primary infection serious risk to fetus. The severe anemia may result in
 - Hydropsfetalis
 - Fetal death before 20th week of pregnancy
 - No evidence of physical deformities in fetus

Diagnosis:

- ✳ Specimens: Serum (principal specimen), Tissue biopsy
 - Virus Detection, Serologic tests, Molecular technique

Treatment:

- ✳ No specific treatment, except intravenous administration of human Ig in cases of persistent infection in immunocompromised patient. Currently, there is no vaccine for B19.
- ✳ Prevention and control involves isolation of susceptible individuals (if possible)

Rubella Virus (German measles or three days measles)

- ✳ Toga viridae (but are not ARBO viruses like the other members), ss +ve sense RNA
- ✳ Icosahedral capsid & Enveloped
- ✳ Spread by respiratory droplets.
- ✳ Rubella Infection can be Postnatal rubella or Congenital Rubella

Postnatal rubella

- ✱ Not serious (sub clinical)but common
- ✱ Acquired via respiratory route
- ✱ IP 2-3 weeks

Pathogenesis

- ✱ Virus →mucosa URT/upper respiratory tract → replication → Cervical LN → Viremia→Antibodies appear about 13-15 days(rash appears at the same time as antibodies)
- ✱ Virus persists only on nasopharynx

Symptoms : sore throat, runny nose, cough, fever, rash (starts on face & goes to trunk and extremities and lasts 12hour to 3-5 days), rash may not always be seen, arthralgia & arthritis(especially in adult women),lymphadenopathy

Although rare complications include: Encephalitis, Thrombocytopenic purpura, Embryopathy (viruria, up to 1 year infectious)

Risks of rubella infection during pregnancy – Very Important congenital Infection

- ✱ Preconception minimal risk
- ✱ 0-12 weeks - 100% risk for fetus & spontaneous abortion occurs in 20% of cases.
- ✱ 13-16 weeks - deafness and retinopathy
- ✱ After 16 weeks - normal development, slight risk of deafness and retinopathy

Congenital Rubella Syndrome

- ✱ Maternal viremia→infection of placenta and fetus →mitosis of infected cells reduced →deranged and hypo-plastic organ development →structural anomalies in newborn.
- ✱ Time of fetal infection very important (EARLIER IN PREGNANCY INFECTION OCCURS GREATER DAMAGE TO FETUS).

Outcome

- ✱ Normal independent lives (Normal survival) or
- ✱ Baby with low body weight,thrombocytopenia, hepatomegaly or
- ✱ Congenital abnormalities
- ✱ Cataracts, micro-ophthalmia, heart defects, deafness, mental retardation etc...
- ✱ Once fetus infected the only effective way is to terminate the pregnancy

Prevention

- ✱ Antenatal screening & rubella vaccination for non-immune women in the immediate postpartum period.
- ✱ Rubella vaccine: live attenuated vaccine in 2 doses

- ✳ Screening women for immunity to rubella prior to pregnancy (indicated by IgG antibodies) - if necessary vaccination.

Laboratory Diagnosis

- ✳ Rising titers of antibody (mainly IgG)
- ✳ Presence of rubella-specific IgM- IgM antibodies do not cross placenta so indicates they are synthesized by fetus.

Treatment

- ✳ No specific treatment
- ✳ Immune globulin is ONLY PREVENTION against infection, no prevention of maternal viremia (discarded).
- ✳ Most important to immunize women of childbearing age and all school-age children.
- ✳ VACCINE NEVER TO PREGNANT WOMEN OR WHO MAY BECOME PREGNANT WITHIN 3 MONTHS OF VACCINATION.

Cytomegalo Virus (Human Herpes Virus Type 5)

- ✳ DNA virus with icosahedral capsid; is the largest of the herpes viruses
- ✳ Infection is common but mostly asymptomatic; symptoms are mostly seen in people with immune defect (T cell defect, pregnancy, infants)
- ✳ Normal individuals - primary infection is usually asymptomatic, occasionally an infectious mononucleosis like illness may be seen. Reactivation is common.
- ✳ Congenital CMV infection - fetal retardation → fetal death
- ✳ Immunocompromised individuals - both primary and recurrent infection may lead to symptomatic disease. Primary CMV infection is usually more severe than recurrent infection, with the exception of bone marrow transplant recipients, where primary and recurrent infections are just as severe.

Signs & Symptoms:

- ✳ Fever, Pneumonitis (85% mortality rate in absence of treatment), Hepatitis, Gastrointestinal manifestations (like colitis), Encephalopathy, Retinitis, Poor graft function.
- ✳ Retinitis (which leads to blindness unless treated), colitis & encephalopathy are the most common symptoms of CMV in AIDS patients
 - CMV is the most common cause of morbidity & mortality in solid organ transplant recipients. It causes fever, pneumonitis, colitis, hepatitis, and poor graft function 1-3 months following transplant.

Congenital CMV Infection

- From mother infected in pregnancy; Maternal infection usually asymptomatic while fetal infection can be asymptomatic to severe and disseminated
- CMV is the most common viral cause of mental retardation
- Fetus damaged at any stage (severe developmental defects, mental retardation, deafness).

Perinatal CMV Infection

- Infected birth canal/breast milk or other secretions.
- Protracted interstitial pneumonitis, Poor weight gain, adenopathy, rash.
- Hepatitis and anemia which persist for months to years

Laboratory Diagnosis

- Cytology/histology –large cytomegalic cells with intranuclear inclusions-“owl’s eye”
- Culture-Gold standard
- Nucleic acid antigen detection, Serology, PCR

Treatment:

- ✱ Ganciclovir, Foscarnet, Cidofovir (HPMCC), Fomivirsen, CMV hyperimmune globulin (effective against CMV pneumonitis).
- Refer to STDs for herpes simplex, syphilis & HBV

BENIGN AND MALIGNANT BREAST LESIONS

- Breast tissue consists of two major structures (ducts and lobules), two types of epithelial cells (luminal and myoepithelial), and two types of stroma (interlobular and intralobular). Successive branching of the large ducts eventually leads to the terminal duct lobular unit.

Benign Epithelial Lesions

Fibrocystic changes

- Fibrocystic changes are the most common breast abnormality seen in premenopausal women. It is associated with hormonal imbalance (increased estrogen to progesterone ratio).
- Fibrocystic changes can be subdivided into non proliferative and proliferative patterns.

Non proliferative Changes

Cysts and Fibrosis

- Cysts are usually multifocal and bilateral. The cysts are called **Blue dome cysts**, because they appear brown to blue. Cysts are lined either by a flattened atrophic epithelium or by metaplastic apocrine cells, **apocrine metaplasia**. Calcifications are common.
- Cysts frequently rupture, releasing secretory material into the adjacent stroma. The resulting chronic inflammation and fibrosis contribute to the palpable nodularity of the breast.

Proliferative Breast Disease Without Atypia

Epithelial Hyperplasia

- Epithelial hyperplasia is recognized by the presence of more than two cell layers. The spectrum of epithelial hyperplasia ranges from mild and orderly to atypical hyperplasias with features that resemble those of in situ carcinoma.
- Sometimes, the proliferating epithelium projects as multiple small papillary excrescences into the ductal lumen (ductal papillomatosis).

Sclerosing Adenosis

- These lesions contain marked intralobular fibrosis and proliferation of small ductules and acini. Grossly, the lesion has a hard, rubbery consistency, similar to that of breast cancer. Histologic examination shows a characteristic proliferation of luminal spaces (adenosis) lined by epithelial cells and myoepithelial cells, yielding masses of small glands within a fibrous stroma.

Complex Sclerosing Lesion

- Have components of sclerosing adenosis, papillomas and epithelial hyperplasia.

Papilloma

- Papillomas grow within a dilated duct and are composed of multiple branching fibrovascular core. Epithelial hyperplasia and apocrine metaplasia are frequently present. Large duct papillomas are situated in the lactiferous sinuses of the nipple and are usually solitary. Small duct papillomas are commonly multiple and located deeper within the ductal system.
- More than 80% of large duct papillomas produce a nipple discharge.

Proliferative Breast Disease with Atypia

- Atypical hyperplasia is a clonal proliferation having some, but not all, of the histologic features that are required for the diagnosis of carcinoma in situ.
- It includes two forms, atypical ductal hyperplasia and atypical lobular hyperplasia. Atypical ductal hyperplasia is distinguished from ductal carcinoma in situ in that it only partially fills involved ducts.
- Atypical lobular hyperplasia consists of cells identical to those of lobular carcinoma in situ, but the cells do not fill or distend more than 50% of the acini within a lobule.

Stromal tumors

Fibroadenoma

- Fibroadenoma is by far the most common benign neoplasm of the female breast. It is a biphasic tumor composed of fibroblastic stroma and epithelium-lined glands. Most occur in women in their 20s and 30s, and they are frequently multiple and bilateral. The tumor is well circumscribed and freely mobile.
- An absolute or relative increase in estrogen is thought to contribute to their development. In addition, fibroadenomas may enlarge late in the menstrual cycle and during pregnancy; after menopause, they may regress and calcify.

Phyllodes Tumor

- Phyllodes tumors, like fibroadenomas, arise from intralobular stroma. Phyllodes tumors are distinguished from fibroadenomas on the basis of higher cellularity, higher mitotic rate, nuclear pleomorphism, stromal overgrowth, and infiltrative borders.

Lesions of Interlobular Stroma

- Tumors of interlobular stroma are composed of stromal cells without an epithelial component. These include pseudoangiomatous stromal hyperplasia and fibromatosis (fibroblasts and myofibroblasts), myofibroblastoma (myofibroblasts), and lipomas.

Malignant Tumors of Interlobular Stroma

- Malignant stromal tumors include angiosarcoma, rhabdomyosarcoma, liposarcoma, leiomyosarcoma, chondrosarcoma and osteosarcoma.

Types of Breast Carcinoma

Carcinoma in Situ

- Carcinoma in situ refers to a neoplastic proliferation of epithelial cells that is confined to ducts and lobules by the basement membrane.

Ductal Carcinoma in Situ (DCIS)

- Most are detected mammographically, because of calcifications.
- DCIS can be divided into two major architectural subtypes, comedo and noncomedo.
- **Comedo DCIS:** is defined by two features: (1) tumor cells with pleomorphic, highgrade nuclei and (2) areas of central necrosis.
- **Noncomedo DCIS:** lacks either high-grade nuclei or central necrosis.

Paget disease: is caused by the extension of DCIS up the lactiferous ducts and into the contiguous skin of the nipple, producing a unilateral crusting exudate over the nipple and areolar skin.

Lobular Carcinoma in Situ

- LCIS is a clonal proliferation of cells within ducts and lobules that grow in a discohesive fashion, usually due to an acquired loss of the tumor suppressive adhesion protein E-cadherin. It does not induce calcifications. It often contains intracellular mucin forming signet ring cells. Most express **ER and PR**.
- LCIS is bilateral in 20% to 40% of cases.

Invasive (Infiltrating) Carcinoma

Invasive ductal carcinoma

- It is usually scirrhus (dense fibrous tissue), firm, poorly circumscribed and hard.
- Larger tumors have hemorrhage, necrosis and cystic degeneration.

Invasive lobular carcinoma

- It consists of cells morphologically identical to the cells of LCIS. The cells invade individually into stroma and are often aligned in “**single-file**” strands or chains, due to mutation that abrogate the function of E-cadherin. Lobular carcinomas are more frequently multicentric and bilateral. These often have a signet ring appearance, and desmoplasia is typically minimal.

Medullary carcinoma

- Two thirds exhibit hypermethylation of the BRCA1 promoter.

- Tumors are characteristically soft and fleshy with a pushing border and little desmoplastic response. It contains a lymphoplasmacytic infiltrate.

Mucinous (Colloid) Carcinomas

- These are slow-growing well-differentiated ER-positive tumors. Nodal metastases are uncommon.

Tubular Carcinoma

- Tubular carcinomas are typically detected as small irregular mammographic densities in women in their forties; they may be multifocal and/or bilateral.
- They have an excellent prognosis.

Male breast

- **Gynecomastia** : Due to imbalance between estrogen and androgen. Histologically, there is ductal epithelial and stromal hyperplasia; lobule formation is rare.
- **Carcinoma**: strongly associated with BRCA2 mutation. Because of the scant amount of male breast tissue, carcinomas tend to invade skin and chest wall earlier and present at higher stage.

BENIGN AND MALIGNANT CERVICAL LESIONS

- Most tumors of the cervix are of epithelial origin and are caused by oncogenic strains of human papillomavirus (HPV). HPV genome encodes two potent oncoproteins called E6 and E7 which inactivate two critical tumor suppressors, p53 and Rb, respectively.

Cervical Intraepithelial Neoplasia (CIN)

- It is also called Squamous Intraepithelial Lesions (SIL).
- SIL is a precancerous epithelial change which usually precedes the development of cancer by many years. It is detected through Pap smear.
- SIL is classified as low-grade or high-grade squamous intraepithelial lesions (LSIL and HSIL, respectively).
- LSIL show only mild dysplasia. The atypia is confined to the basal third of the epithelium and koilocytotic change in the superficial layers of the epithelium. LSIL was previously known as **CIN I**.

- HSIL exhibit moderate to severe dysplasia and involve progressively more of the epithelial thickness; this category also includes carcinoma in situ. The atypia extends to two thirds (or more) of the epithelial thickness. HSIL represents the former **CIN II and CIN III**.

Cervical Carcinoma

- The most common cervical carcinomas are squamous cell carcinomas (75%), followed by adenocarcinomas and mixed adenosquamous carcinomas (20%) and small cell neuroendocrine carcinomas (less than 5%). All of these types of carcinomas are caused by HPV.
- Grossly: Lesions can be exophytic or infiltrative.
- Microscopically: Squamous lesions can be keratinizing or nonkeratinizing; adenocarcinomas tend to be glandular but relatively mucin depleted; adenosquamous lesions are composed of intermixed malignant squamous and glandular elements; neuroendocrine tumors resemble small cell malignancy of the lung.

Endocervical Polyp

- Endocervical polyps are benign polypoid masses seen protruding from the endocervical mucosa (sometimes through the exocervix).

PELVIC INFLAMMATORY DISEASE

- Pelvic inflammatory disease (PID) is an infection that begins in the vulva or vagina and spreads upward to involve most of the structures in the female genital system: such as endometritis, salpingitis, oophoritis, myometritis, parametritis & infection in the pelvic peritoneum
- ***Neisseria gonorrhea* and *chlamydia*** infections are the main causes of PID. Infections after spontaneous or induced abortions and normal or abnormal deliveries (called puerperal infections) are also important causes of PID. In these situations the infections are typically polymicrobial.

PATHOLOGY OF PREGNANCY

1. Disorders of Early Pregnancy

Spontaneous Abortion

- Spontaneous abortion (“miscarriage”) is defined as pregnancy loss before 20 weeks’ gestation.

Ectopic Pregnancy

- Ectopic pregnancy denotes embryo implantation at a site other than the uterus, often in the fallopian tubes (90%) but also in the ovary or abdominal cavity.

2. Disorders of Late Pregnancy

Twin Placentas

- Twin pregnancies arise from fertilization of two ova (i.e., dizygotic) or division of one fertilized ovum (i.e., monozygotic). There are three basic types of twin placentas : diamnionic dichorionic (which may be fused), diamnionic monochorionic, and monoamnionic monochorionic.
- One complication of monochorionic twin pregnancy is **twin-twin transfusion syndrome**. It is imbalanced blood flow through an arteriovenous shunt and the subsequent disparities in blood volume can lead to death of one or both fetuses.

Abnormalities of Placental Implantation

- **Placenta previa**: is a condition in which the placenta implants in the lower uterine segment or cervix, often leading to serious third trimester bleeding.
- **Placenta accrete**: occurs when there is absence of decidua and the placenta adheres directly to the myometrium; at delivery, the placenta fails to separate.

Placental Infections

- **Ascending** (usually bacterial) infections via the birth canal are most common; these can cause infection of the chorionic membranes (**acute chorioamnionitis**).
- **Hematogenous infections** can result from a maternal septicemia. These are characterized by villous chronic inflammation (villitis).

Preeclampsia and Eclampsia

- **Preeclampsia** is a syndrome characterized by hypertension, proteinuria, and edema. Approximately 10% of women with severe preeclampsia develop microangiopathic **Hemolysis**, **Elevated Liver enzymes**, and **Low Platelets**, referred to as the **HELLP syndrome**.
- **Eclampsia** is a more severe form associated with seizures and coma.

3. Gestational Trophoblastic Disease

Hydatidiform Mole

- Hydatidiform moles are characterized by cystic swelling of chorionic villi, accompanied by variable trophoblastic proliferation; **these can be precursors of choriocarcinoma**. It presents with grape like structures consisting of **swollen edematous (hydropic) villi**. It is classified as **complete and partial mole**.

Complete mole: occurs when an egg that has lost its chromosomes is fertilized by 1 or 2 sperm; all genetic material is therefore paternally derived. About 90% of complete moles derive from the duplicated genetic material of one sperm and are 46,XX; the remainder derive from two sperm and are 46, XX or 46,XY.

- In complete mole, the microscopic abnormalities involve all or most of the villous tissue. hCG levels greatly exceed those of a normal pregnancy of similar gestational age.
- It has Increased risk of choriocarcinoma when compared to partial mole.

Partial mole: occurs when an egg with normal chromosomal content is fertilized by two sperm. The karyotype is 69,XXX or 69,XXY. There is no increased risk of choriocarcinoma. Partial moles show only focal edema and focal and slight trophoblastic proliferation.

- Serum hCG is less elevated.

Invasive Mole

- An invasive mole penetrates and may even perforate the uterine wall, associated with proliferating cytotrophoblasts and syncytiotrophoblasts; villi can embolize to distant sites but do not grow.

Choriocarcinoma

- Lesions consist of mixed cytotrophoblast and syncytiotrophoblast proliferations. hCG titers are elevated. The tumor invades the underlying endometrium, penetrates blood vessels and lymphatics, and can metastasize widely. **Gestational choriocarcinomas are highly sensitive to chemotherapy.**

Placental-Site Trophoblastic Tumor

- It represents neoplastic proliferation of extravillous (intermediate) trophoblasts. The lesion differs from choriocarcinoma because syncytio- and cytotrophoblastic elements are absent and the tumors make lower levels of hCG.

DISEASES OF PROSTATE

- In the adult, prostatic parenchyma can be divided into four biologically and anatomically distinct zones or regions: the **peripheral, central, transitional, and periurethral zones**.

Inflammation

- **Acute bacterial prostatitis:** This is typically caused by organisms associated with UTI. Prostatic infection occurs through urinary reflux or lymphohematogenous seeding from more distant sites; it can also follow catheterization or surgical manipulation.
- **Chronic bacterial prostatitis:** can be asymptomatic or associated with low back pain, suprapubic and perineal discomfort, and dysuria. It is frequently associated with a history of recurrent UTI, but without previous acute prostatitis.
- **Chronic abacterial prostatitis:** This is the most common form of prostatitis.
- **Granulomatous prostatitis**

Benign Enlargement

Benign Prostatic Hyperplasia or Nodular Hyperplasia

- BPH or nodular hyperplasia is an extremely common disorder caused by periurethral epithelial and stromal hyperplasia that compresses the urethra; symptoms are related to urinary flow obstruction. The critical mediator in this process is dihydrotestosterone (DHT).
- The gland is enlarged by nodules primarily in the transitional and periurethral zones.
- Nodules are composed of variable mixtures of proliferating glands and fibromuscular stroma when seen microscopically.

Tumors

Adenocarcinoma

- Prostatic carcinoma is the most common form of cancer in men. It is uncommon in Asians, and is more common in blacks than in whites. It mainly occurs in the older age group.

Pathogenesis:

- **Androgens:** Prostate cancer cells depend on androgen interactions with AR to activate pro-growth and pro-survival genes. Castration and anti-androgen therapies slow tumor progression.

- **Germline genes:** Risk increases with the number of first-degree relatives with prostate cancer. BRCA2 mutations increase risk 20-fold.

Most cases arise in the peripheral zone of the prostate, usually in the posterior prostate. Most are **well-differentiated adenocarcinomas**.

The Gleason system stratifies prostate cancers into five grades based on their glandular patterns (i.e., 1=closest to normal; 5=no glandular differentiation), without regard to cytologic features.

Prostate cancer is associated at an early stage with an increase in serum **prostate-specific antigen (PSA)**. Diagnosis is most often by **rectal examination**.

Thyroid Hormones in Therapeutics and Anti-thyroid Hormones

- The thyroid gland facilitates normal growth and maturation by maintaining a level of metabolism in the tissues. The two major thyroid hormones are triiodothyronine (T3); the most active form) and thyroxine (T4). The iodine necessary for the synthesis of these molecules comes from food or iodide supplements.
- Thyroid function is controlled by a tropic hormone, thyroid-stimulating hormone (TSH; thyrotropin) which is in turn governed by hypothalamic thyrotropin releasing hormone (TRH).
- Hypothyroidism results in bradycardia, poor resistance to cold, and mental and physical slowing (in children, this can cause mental retardation and dwarfism). But excess of thyroid hormones (hyperthyroidism) causes then tachycardia and cardiac arrhythmias, body wasting, nervousness, tremor, and excess heat production can occur.
- Both T4 and T3 must dissociate from thyroxine-binding plasma proteins prior to entry into cells, either by diffusion or by active transport. Only about 0.04% of total T4 and 0.4% of T3 exist in the free form.
- The primary pathway for the peripheral metabolism of thyroxine is deiodination.

Pharmacokinetics

- Thyroxine is absorbed best in the duodenum and ileum. Oral bioavailability of current thyroxine (T4) averages 80% but T3 is almost completely absorbed (95%). For parenteral use, the intravenous route is preferred for both hormones.
- Drugs that induce hepatic microsomal enzymes (eg, rifampin) increase the metabolism of both T4 and T3. In patients with hyperthyroidism, the metabolic clearances of T4 and T3 are increased and the half-lives decreased; the opposite is true in patients with hypothyroidism.

Thyroid Preparations

- These preparations may be synthetic (levothyroxine, liothyronine, liotrix) or of animal origin (desiccated thyroid).

- **Levothyroxine (T4):** is the preparation of choice for thyroid replacement and suppression therapy because of its stability.
 - It is low cost, lack of allergenic foreign protein and long half-life.
 - Since T4 is converted to T3 intracellularly, thus administration of levothyroxine produces both T4 and T3.
- **Liothyronine (T3):** is faster acting but has a shorter half-life and is more expensive.
 - Three to four times more potent than levothyroxine.
 - Not recommended for routine replacement therapy because of its shorter half-life (24 hours).
 - Greater hormone activity and risk of cardiotoxicity.

Anti-thyroid Agents

- Reduction of thyroid activity and hormone effects can be accomplished by agents that interfere with the production of thyroid hormones, by agents that modify the tissue response to thyroid hormones, or by glandular destruction with radiation or surgery.
- **Thioamides**
 - Methimazole and propylthiouracil (PTU) are small sulfur-containing thioamides that inhibit thyroid hormone synthesis by blocking peroxidase-catalyzed reactions, iodination of the tyrosine residues of thyroglobulin, and coupling of DIT and MIT.
 - Propylthiouracil and, to a much lesser extent, methimazole inhibit peripheral conversion of T4 to T3.
 - Do not inhibit the release of preformed thyroid hormone → onset of activity is usually slow.
 - Methimazole is generally preferred because it can be administered once per day.
 - PTU is preferred in pregnancy.
 - Skin rash (common) and severe reactions (rare) such as vasculitis, agranulocytosis, hypoprothrombinemia, and liver dysfunction.
- **Iodides**
 - Iodide salts inhibit iodination of tyrosine and thyroid hormone release. Decrease the size and vascularity of the hyperplastic thyroid gland.
 - Their onset of action occurs rapidly, within 2–7 d.
 - Transient effects → gland “escapes” from iodide block after several weeks.
 - Used in the management of thyroid storm and to prepare patients for surgical resection of a hyperactive thyroid.
 - Usual forms: Lugol’s solution and saturated solution of potassium iodide.
 - Rash, drug fever, metallic taste, bleeding disorders, and, rarely, anaphylactic reactions.
- **Radioactive Iodine**
 - Taken up and concentrated in the gland → severely damage the gland without endangering other tissues.
 - Effective dose of ^{131}I can produce a permanent cure of thyrotoxicosis without surgery.
 - Not used with pregnant and nursing women.
- **Anion Inhibitors**
 - Thiocyanate (SCN^-) and perchlorate (ClO_4^-) block the uptake of iodide → through competitive inhibition of the iodide transporter.

- However, potassium perchlorate is rarely used clinically because it is associated with aplastic anemia.
- **Other Drugs: β Blockers**
 - These agents are particularly useful in controlling the tachycardia and other cardiac abnormalities of severe thyrotoxicosis.
 - Propranolol also inhibits the peripheral conversion of T4 to T3.

Anti-diabetic Drugs

- There are two types of diabetes mellitus called Type I and Type II.
- The clinical history and course of these 2 forms differ considerably, but treatment in both cases requires careful attention to diet, fasting and postprandial blood glucose concentrations, and serum concentrations of hemoglobin A1c, a glycosylated hemoglobin that serves as a marker of glycemia.
- Type 1 diabetes requires treatment with insulin. The early stages of type 2 diabetes usually can be controlled with noninsulin anti-diabetic drugs. But insulin is required in later stages.
- Insulin is synthesized as the prohormone proinsulin. Insulin has important effects on almost every tissue of the body.
- Insulin increases the storage of glucose as glycogen in the liver. Insulin stimulates glycogen synthesis and protein synthesis in skeletal muscle. Insulin facilitates triglyceride storage by activating plasma lipoprotein lipase, increasing glucose transport into cells and reduce intracellular lipolysis.

Insulin Preparations

- Manufactured by bacterial recombinant DNA technology.
- The goals of insulin therapy are to control both basal and postprandial (after a meal) glucose levels while minimizing the risk of hypoglycemia.
- **Rapid Acting (Insulin Lispro, Insulin Aspart, and Insulin Glulisine)**
 - Have rapid onsets and early peaks of activity that permit control of postprandial glucose levels.
 - Small amino acid changes in primary sequences → rapid entry into circulation but doesn't affect interaction with insulin receptor.
 - The rapid-acting insulins are injected immediately before a meal → preferred for continuous subcutaneous infusions. Also for emergency treatment in uncomplicated DKA.
 - These agents offer more flexible treatment regimens and may lower the risk of hypoglycemia.
 - Peak levels of insulin lispro are seen at 30 to 90 minutes after injection. Has shorter duration of action.
 - Usually not used alone but with a longer-acting insulin to ensure proper glucose control.
- **Short Acting (Regular Insulin)**

- Used intravenously in emergencies. Administered subcutaneously in ordinary maintenance regimens alone or mixed with intermediate or long acting preparations.
- Administered 1h or more before a meal.
- Regular insulin, insulin lispro, and insulin aspart are pregnancy category B, and insulin glulisine is pregnancy category C.
- Peak levels after 50 to 120 minutes.
- **Intermediate Acting (Neutral Protamine Hagedorn or NPH Insulin)**
 - A combination of regular insulin and protamine.
 - Delayed onset and peak of action. Its duration of action is intermediate because of the delayed absorption from its conjugation with protamine, forming a less-soluble complex.
 - NPH insulin is often combined with regular and rapid-acting insulins.
 - NPH insulin should only be given subcutaneously (never IV) and is useful in treating all forms of diabetes except diabetic ketoacidosis and emergency hyperglycemia.
- **Long Acting (Insulin Glargine and Insulin Detemir)**
 - Provide a peakless basal insulin level lasting more than 20h. Helps control basal glucose levels without producing hypoglycemia.
 - The isoelectric point of insulin glargine is lower than that of human insulin, leading to precipitation at the injection site and extending its action.
 - Insulin detemir has a fatty-acid side chain. This addition enhances association to albumin. Slow dissociation from albumin results in long-acting properties.
 - Must be given subcutaneously. Neither insulin detemir nor insulin glargine should be mixed in the same syringe with other insulins → alter properties.
- The most common complication is hypoglycemia, resulting from excessive insulin effect. To prevent the brain damage that may result from hypoglycemia, prompt administration of glucose or of glucagon is essential.
- With the current use of highly purified human insulins, immunologic complications are uncommon.

Insulin Secretagogues: Sulfonylureas

- Stimulate the release of endogenous insulin by promoting closure of potassium channels in the pancreatic B-cell membrane. Not effective in patients who lack functional pancreatic B cells.
- Reduction in hepatic glucose production. Increase in peripheral insulin sensitivity.
- **Tolbutamide**
 - Well absorbed but rapidly metabolized in the liver. Its duration of effect is relatively short, with an elimination half-life of 4–5 hours.
 - Prolonged hypoglycemia in patients receiving certain drugs that inhibit the metabolism of tolbutamide.
- **Chlorpropamide**
 - Half-life of 32 hours and is slowly metabolized in the liver.
 - Contraindicated in patients with hepatic or renal insufficiency. Prolonged hypoglycemic reactions are more common in elderly patients.
- **Tolazamide**

- Comparable to chlorpropamide in potency but has a shorter duration of action. More slowly absorbed.
- Tolazamide is metabolized to several compounds that retain hypoglycemic effects.
- **Glyburide**
 - Metabolized in the liver. Contraindicated in the presence of hepatic impairment and in patients with renal insufficiency.
 - Has minimal transfer across the placenta and may be a reasonably safe alternative to insulin therapy for diabetes in pregnancy.
- **Glipizide**
 - Has the shortest half-life (2–4 hours). The regular formulation of glipizide is much less likely than glyburide to produce serious hypoglycemia.
 - At least 90% of glipizide is metabolized in the liver to inactive products, and 10% is excreted unchanged in the urine. Contraindicated in patients with significant hepatic or renal impairment → severe hypoglycemia.
- **Glimepiride**
 - Once-daily use as monotherapy or in combination with insulin. Has a long duration of effect.
 - Achieves blood glucose lowering with the lowest dose of any sulfonylurea compound.
 - It is completely metabolized by the liver to metabolites with weak or no activity.
- Shortcomings of the sulfonylureas are their propensity to cause weight gain, hyperinsulinemia, and hypoglycemia.

Insulin Secretagogues: Glinides

- Includes Repaglinide and Nateglinide. Although they are not sulfonylureas, they have common actions.
- They bind to a distinct site on the sulfonylurea receptor of ATP-sensitive potassium channels, thereby initiating a series of reactions culminating in the release of insulin.
- Have a rapid onset and a short duration of action. They are particularly effective in the early release of insulin that occurs after a meal and are categorized as postprandial glucose regulators.
- Combined therapy of these agents with metformin or the glitazones has been shown to be better than monotherapy. Glinides should not be used in combination with sulfonylureas due to overlapping mechanisms of action.
- These drugs are well absorbed orally after being taken 1 to 30 minutes before meals.
- Metabolized to inactive products by the liver and are excreted through the bile. Enzyme inducers and inhibitors may interfere with their glucose lowering effect.
- These agents must be used with caution in patients with hepatic impairment.

Biguanides

- **Metformin**
 - The primary member of the biguanide group, reduces postprandial and fasting glucose levels and classified as insulin sensitizer. It increases glucose uptake and use by target tissues, thereby decreasing insulin resistance. Other effects include stimulation of glucose

uptake and glycolysis in peripheral tissues, slowing of glucose absorption from the gastrointestinal tract, and reduction of plasma glucagon levels.

- Biguanides inhibit hepatic and renal gluconeogenesis (involve activation of an AMP-stimulated protein kinase).
- Reduces endogenous insulin production presumably through enhanced insulin sensitivity → insulin sparing effect. Increasingly the drug of first choice in overweight patients with type 2 diabetes.
- Unlike the sulfonylureas, the biguanides do not cause hypoglycemia.
- Their most common toxicity is gastrointestinal distress (nausea, diarrhea), and they can cause lactic acidosis.

Thiazolidinediones

- Another group of agents that are insulin sensitizers also called the glitazones.
- Increase target tissue sensitivity to insulin by activating the peroxisome proliferator-activated receptor- γ nuclear receptor (PPAR- γ receptor). Ligands for PPAR γ regulate adipocyte production and secretion of fatty acids as well as glucose metabolism, resulting in increased insulin sensitivity in adipose tissue, liver, and skeletal muscle. A primary effect of the thiazolidinediones is increasing glucose uptake in muscle and adipose tissue.
- Although insulin is required for their action, these drugs do not promote its release from the pancreatic β cells, so hyperinsulinemia is not a risk.
- They also inhibit hepatic gluconeogenesis and have effects on lipid metabolism and the distribution of body fat. Thiazolidinediones reduce both fasting and postprandial hyperglycemia.
- They are used as monotherapy or in combination with insulin or other oral antidiabetic drugs.
- The two members of this class currently available are pioglitazone and rosiglitazone.
- Well absorbed after oral administration and are extensively bound to serum albumin. Both undergo extensive metabolism by different CYP450 isozymes.
- When these drugs are used alone, hypoglycemia is extremely rare. Because there have been deaths from hepatotoxicity in patients who took troglitazone (the first thiazolidinedione) it is recommended that liver enzyme levels of patients on these medications be measured initially and periodically thereafter. Very few cases of liver toxicity have been reported with rosiglitazone or pioglitazone.
- Can cause fluid retention, which presents as mild anemia and edema and may increase the risk of heart failure. Several meta-analyses identified a potential increased risk of myocardial infarction and death from cardiovascular causes with rosiglitazone.
- Pioglitazone and troglitazone induce cytochrome P450 activity (especially the 3A4 isozyme) and can reduce the serum concentrations of drugs that are metabolized by these enzymes (eg, oral contraceptives, cyclosporine).

α -Glucosidase Inhibitors

- Acarbose and miglitol are orally active drugs used for the treatment of patients with type 2 diabetes. These drugs are taken at the beginning of meals.
- carbohydrate analogs that act within the intestine to inhibit α -glucosidase, an enzyme necessary for the conversion of complex starches, oligosaccharides, and disaccharides to the monosaccharides that can be transported out of the intestinal lumen and into the

bloodstream. As a result of slowed absorption, postprandial hyperglycemia is reduced. These drugs lack an effect on fasting blood sugar.

- The primary adverse effects of the α -glucosidase inhibitors include flatulence, diarrhea, and abdominal pain resulting from increased fermentation of unabsorbed carbohydrate by bacteria in the colon. Patients with inflammatory bowel disease, colonic ulceration, or intestinal obstruction should not use these drugs.
- Patients taking an α -glucosidase inhibitor who experience hypoglycemia should be treated with oral glucose (dextrose) and not sucrose, because the absorption of sucrose will be delayed.

Synthetic Amylin Analogs

- Pramlintide is a synthetic amylin analog that is indicated as an adjunct to mealtime insulin therapy in patients with type 1 and type 2 diabetes. Amylin contributes to glycemic control by activating high-affinity receptors that are a complex of the calcitonin receptor and a receptor activity modifying receptor (RANK).
- Pramlintide suppresses glucagon release, slows gastric emptying, and works in the CNS to reduce appetite (improve satiety).
- Administered by subcutaneous injection and should be injected immediately prior to meals. When pramlintide is initiated, the dose of rapid- or short-acting insulin should be decreased by 50 percent prior to meals to avoid a risk of severe hypoglycemia.
- Adverse effects are mainly gastrointestinal and consist of nausea, anorexia, and vomiting.

Incretin Mimetics

- Glucagon-like peptide-1 (GLP-1) is a member of the incretin family of peptide hormones, which are released from endocrine cells in the epithelium of the bowel in response to food. Incretin hormones are responsible for 60 to 70 percent of postprandial insulin secretion.
- Exenatide and liraglutide are injectable incretin mimetics used for the treatment of patients with type 2 diabetes. These agents not only improve glucose-dependent insulin secretion but also slow gastric emptying time, decrease food intake, decrease postprandial glucagon secretion, and promote β -cell proliferation. Consequently, weight gain and postprandial hyperglycemia are reduced, and HbA_{1c} levels decline.
- Being polypeptides, exenatide and liraglutide must be administered subcutaneously. Liraglutide is long acting and can be used as once daily dosing. Exenatide is eliminated by glomerular filtration and is cleared rapidly so twice daily injection is needed 60 minutes before dinner and breakfast. Dose regulation for exenatide in renal impairment.
- The major adverse effects are gastrointestinal disturbances, particularly nausea during initial therapy, and hypoglycemia when exenatide is combined with a sulfonylurea. The drug has also caused serious and sometimes fatal acute pancreatitis.

Dipeptidyl Peptidase-IV Inhibitor

- Sitagliptin and saxagliptin are orally active dipeptidyl peptidase-IV (DPP-IV) inhibitors used for the treatment of patients with type 2 diabetes.
- These drugs inhibit the enzyme DPP-IV, which is responsible for the inactivation of incretin hormones such as glucagon-like peptide-1 (GLP-1). Prolonging the activity of incretin hormones

results in increased insulin release in response to meals and a reduction in inappropriate secretion of glucagon.

- In general, DPP-IV inhibitors are well tolerated, with the most common adverse effects being nasopharyngitis and headache.

Adrenocortical Hormones in Therapeutics

- The corticosteroids are steroid hormones produced by the adrenal cortex. They consist of 2 major physiologic and pharmacologic groups:
 - Glucocorticoids, which have important effects on intermediary metabolism, catabolism, immune responses, and inflammation.
 - Mineralocorticoids, which regulate sodium and potassium reabsorption in the collecting tubules of the kidney.

Glucocorticoids

- Cortisol is the principal human glucocorticoid. Normally, its production is diurnal, with a peak early in the morning followed by a decline and then a secondary, smaller peak in the late afternoon. Factors such as stress and levels of the circulating steroid influence secretion. The effects of cortisol are many and diverse.
 - Metabolic Effects
 - Glucocorticoids stimulate gluconeogenesis. They stimulate protein catabolism (except in the liver) and lipolysis, thereby providing the building blocks and energy that are needed for glucose synthesis.
 - Immunosuppressive Effects
 - Glucocorticoids inhibit cell-mediated immunologic functions, especially those dependent on lymphocytes. These agents are actively lymphotoxic and, as such, are important in the treatment of hematologic cancers.
 - The drugs do not interfere with the development of normal acquired immunity but delay rejection reactions in patients with organ transplants.
 - Anti-inflammatory Effects
 - Glucocorticoids cause a decrease in eosinophils, basophils, monocytes, and lymphocytes by redistributing them from the circulation to lymphoid tissue.
 - The most important therapeutic property of the glucocorticoids is their ability to dramatically reduce the inflammatory response and to suppress immunity.
 - Also involved is the indirect inhibition of phospholipase A₂, which blocks the release of arachidonic acid (the precursor of the prostaglandins and leukotrienes) from membrane-bound phospholipid. Cyclooxygenase-2 synthesis in inflammatory cells is further reduced, lowering the availability of prostaglandins. In addition, interference with mast cell degranulation results in decreased histamine and capillary permeability.
 - Effect on Endocrine System

- Feedback inhibition of corticotropin production by elevated glucocorticoids causes inhibition of further synthesis of both glucocorticoid and thyroid-stimulating hormones.
- Effect on other Systems
 - Adequate cortisol levels are essential for normal glomerular filtration. High doses of glucocorticoids stimulate gastric acid and pepsin production and may exacerbate ulcers. Effects on the central nervous system that influence mental status have been identified. Chronic glucocorticoid therapy can cause severe bone loss and myopathy.
 - Effects of corticosteroids on other systems are mostly associated with the adverse effects of the hormones.

Mineralocorticoids

- Mineralocorticoids help to control the body's water volume and concentration of electrolytes, especially sodium and potassium.
- Aldosterone acts on kidney tubules and collecting ducts, causing a reabsorption of sodium, bicarbonate, and water. Conversely, aldosterone decreases reabsorption of potassium, which, with H^+ , is then lost in the urine.

Use in Therapeutics

- Replacement therapy for adrenocortical insufficiency (Addison disease).
 - Primary insufficiency is caused by adrenal cortex dysfunction.
 - Hydrocortisone, which is identical to natural cortisol, is given to correct the deficiency. Administration of fludrocortisone, a potent synthetic mineralocorticoid with some glucocorticoid activity, may also be necessary to raise the mineralocorticoid activity to normal levels.
 - Secondary and tertiary deficiencies are caused by a defect either in CRH production by the hypothalamus or in corticotropin production by the pituitary. Mineralocorticoid synthesis and secretion is less affected.
 - Hydrocortisone is used for treatment of these deficiencies.
- Diagnosis of Cushing syndrome.
 - Cushing syndrome is caused by a hypersecretion of glucocorticoids.
 - Excessive release of corticotropin by the anterior pituitary or an adrenal tumor.
 - The dexamethasone suppression test is used to diagnose and differentiate the cause of Cushing syndrome.
 - Suppresses cortisol release in individuals with pituitary-dependent Cushing syndrome, but it does not suppress glucocorticoid release from adrenal tumors.
- Replacement therapy for congenital adrenal hyperplasia.
 - Resulting from an enzyme defect in the synthesis of one or more of the adrenal steroid hormones.
 - Treatment of this condition requires administration of sufficient corticosteroids to normalize the patient's hormone levels by suppressing release of CRH and ACTH.
- Relief of inflammatory symptoms.

- Glucocorticoids dramatically reduce the manifestations of inflammation (for example, rheumatoid and osteoarthritic inflammation as well as inflammatory condition of the skin), including redness, swelling, heat, and tenderness that are commonly present at the inflammatory site.
- Treatment of allergies.
 - Glucocorticoids are beneficial in the treatment of the symptoms of bronchial asthma; allergic rhinitis; and drug, serum, and transfusion allergic reactions. Triamcinolone and others are applied topically to the respiratory tract through inhalation from a metered-dose dispenser. This minimizes systemic effects and allows the patient to significantly reduce or eliminate the use of oral steroids.
- Acceleration of lung maturation.
 - Fetal cortisol is a regulator of lung maturation. Consequently, a dose of betamethasone or dexamethasone is administered intramuscularly to the mother 48 hours prior to birth, followed by a second dose 24 hours before delivery.
- Prednisone is preferred in pregnancy because it minimizes steroid effects on the fetus. It is a prodrug that is not converted to the active compound, prednisolone in the fetal liver. Any prednisolone formed in the mother is biotransformed to prednisone by placental enzymes.

Adverse Effects

- Osteoporosis is the most common adverse effect due to the ability of glucocorticoids to suppress intestinal Ca^{2+} absorption, inhibit bone formation, and decrease sex hormone synthesis. Patients are advised to take calcium and vitamin D supplements.
- The classic Cushing-like syndrome (that is, redistribution of body fat, puffy face, increased body hair growth, acne, insomnia, and increased appetite) are observed when excess corticosteroids are present.
- Hyperglycemia may develop and lead to diabetes mellitus. Diabetic patients should monitor their blood glucose and adjust their medications accordingly.
- Withdrawal from these drugs can be a serious problem. Abrupt removal of the corticosteroids causes an acute adrenal insufficiency syndrome that can be lethal. Dose must be tapered according to the individual, possibly through trial and error.

Sex Hormones in Therapeutics

- The gonadal hormones include the steroids of the ovary (estrogens and progestins) and testis (chiefly testosterone).

Ovarian Hormones

- The ovary is the primary source of gonadal hormones in women during the childbearing years
- **Estrogens**
 - The major ovarian estrogen in women is estradiol. Estradiol has low oral bioavailability. It can also be administered via transdermal patch, vaginal cream, or intramuscular injection.

- Mixtures of conjugated estrogens from biologic sources (eg, Premarin) are used orally for hormone replacement therapy (HRT). Synthetic estrogens with high bioavailability (eg, ethinyl estradiol, mestranol) are used in hormonal contraceptives.
- Estrogen has many metabolic effects: It modifies serum protein levels and reduces bone resorption. It enhances the coagulability of blood and increases plasma triglyceride levels while reducing low-density lipoprotein (LDL) cholesterol and increasing high-density lipoprotein (HDL) cholesterol.
- Estrogens are used in the treatment of hypogonadism in young females. Another use is as HRT in women with estrogen deficiency resulting from premature ovarian failure, menopause, or surgical removal of the ovaries. It is effective also in preventing bone loss and osteoporosis. The estrogens are components of hormonal contraceptives.
- Nausea and breast tenderness are among the most common adverse effects of estrogen therapy. Postmenopausal uterine bleeding can occur. In addition, the risk of thromboembolic events, myocardial infarction, and breast and endometrial cancer is increased with use of estrogen therapy. The increased risk of endometrial cancer can be offset by including a progestogen along with the estrogen therapy.
- **Selective Estrogen Receptor Modulators**
 - Selective estrogen-receptor modulators (SERMs) are a class of estrogen related compounds that interact at estrogen receptors but have different effects depending on the tissues. This category includes tamoxifen, raloxifene, toremifene (orphan drug), and clomiphene.
 - Considered to be the first SERM, tamoxifen competes with estrogen for binding to the estrogen receptor in breast tissue. Raloxifene is a second-generation SERM that is related to tamoxifen. In addition, raloxifene decreases bone resorption and overall bone turnover. Bone density is increased, and vertebral fractures are decreased.
 - Clomiphene acts as a partial estrogen agonist and interferes with the negative feedback of estrogens on the hypothalamus. This effect thereby increases the secretion of gonadotropin-releasing hormone and gonadotropins leading to stimulating ovulation.
 - Tamoxifen is currently used in the palliative treatment of metastatic breast cancer in postmenopausal women. It may also be used as adjuvant therapy following mastectomy or radiation in breast cancer and as a prophylactic therapy to reduce the risk of breast cancer in high risk patients.
 - Raloxifene is approved for the prophylaxis of breast cancer in high-risk women and also for the prevention and treatment of osteoporosis in postmenopausal women.
 - Clomiphene has been used successfully to treat infertility associated with anovulatory cycles.
 - The most frequent adverse effects of tamoxifen treatment are hot flashes and nausea. Due to its estrogenic activity in the endometrium, hyperplasia and malignancies have been reported in women who have been maintained on tamoxifen.
 - In case of raloxifene there is an increased risk of deep-vein thrombosis, pulmonary embolism, and retinal-vein thrombosis.
- **Progestins**

- Progesterone is the major progestin in humans. A micronized form is used orally for HRT, and progesterone-containing vaginal creams are also available. Synthetic progestins (eg, medroxyprogesterone) have improved oral bioavailability.
- Progesterone induces secretory changes in the endometrium and is required for the maintenance of pregnancy.
- Progestins do not significantly affect plasma proteins, but they do affect carbohydrate metabolism and stimulate the deposition of fat. High doses suppress gonadotropin secretion and often cause anovulation in women.
- The major clinical uses of progestogens are to treat a hormonal deficiency and for contraception.
- For contraception, they are generally used with estrogens, either in combination or in a sequential manner. Progesterone by itself is not used widely as a contraceptive therapy because of its rapid metabolism, resulting in low bioavailability.
- Norethindrone, norethindrone acetate, norgestrel levonorgestrel, desogestrel, norgestimate, and drospirenone are synthetic agents resistant to first-pass metabolism → lower doses when administered orally.
- Medroxyprogesterone acetate is an injectable contraceptive, and the oral form is a common progestin component of postmenopausal hormone therapy.
- Also used for the control of dysfunctional uterine bleeding, treatment of dysmenorrhea, and management of endometriosis and infertility. Progesterone is used in assisted reproductive technology methods to promote and maintain pregnancy
- The major adverse effects associated with the use of progestins are headache, depression, weight gain, and changes in libido. They may increase blood pressure and decrease HDL. Long-term use of high doses in premenopausal women is associated with a reversible decrease in bone density.
- **Antiprogestin**
 - Mifepristone is a progesterone antagonist with partial agonist activity.
 - Its major use is as an abortifacient in early pregnancy (up to 49 d after the last menstrual period). Due to interference with the progesterone needed to maintain pregnancy. Mifepristone is often combined with the prostaglandin analog misoprostol (administered orally or intravaginally) to induce uterine contractions.
 - The major adverse effects are significant uterine bleeding and the possibility of an incomplete abortion.

Androgens

- Testosterone and related androgens are produced in the testis, the adrenal, and, to a small extent, the ovary.
- Because of rapid hepatic metabolism, testosterone given orally has little effect. It may be given by injection in the form of long-acting esters or transdermal patch.
- The major effect of androgenic hormones, in addition to development and maintenance of normal male characteristics, is an anabolic action that involves increased muscle size and strength and increased red blood cell production. Excretion of urea nitrogen is reduced, and nitrogen balance becomes more positive. Testosterone also helps maintain normal bone density.

- The primary clinical use of the androgens is for replacement therapy in hypogonadism. Androgens have also been used to stimulate red blood cell production in certain anemias and to promote weight gain in patients with wasting syndromes (eg, AIDS patients). The anabolic effects have been exploited illicitly by athletes to increase muscle bulk and strength and perhaps enhance athletic performance.
- Use of androgens by females results in virilization (hirsutism, enlarged clitoris, deepened voice) and menstrual irregularity. If pregnant → androgens can cause virilization of the fetus's external genitalia.
- Excessive doses in men can result in feminization (gynecomastia, testicular shrinkage, infertility) → feedback inhibition of pituitary.

Contraceptives

- Hormonal contraceptives contain either a combination of an estrogen and a progestin or a progestin alone. Hormonal contraceptives are available in a variety of preparations, including oral pills, long-acting injections, transdermal patches, vaginal rings, and intrauterine devices (IUDs).
- Postcoital or emergency contraception reduces the probability of pregnancy to between 0.2 and 3 percent. Emergency contraception uses high doses of progestin (for example, levonorgestrel) or high doses of estrogen (ethinyl estradiol) plus progestin (levonorgestrel) administered within 72 hours of unprotected intercourse (the "morning-after pill"). A second dose of emergency contraception should be taken 12 hours after the first dose.
- The combination hormonal contraceptives have several actions, including inhibition of ovulation (the primary action) and effects on the cervical mucus glands, uterine tubes, and endometrium that decrease the likelihood of fertilization and implantation.
- Users of combination hormonal contraceptives have reduced risks of ovarian cysts, ovarian and endometrial cancer, benign breast disease, and pelvic inflammatory disease as well as a lower incidence of ectopic pregnancy, iron deficiency anemia, and rheumatoid arthritis.
- Major adverse effects include:
 - Thromboembolism: relate to the action of the estrogenic component on blood coagulation.
 - The incidence of cervical cancer may be increased with oral contraceptives and early onset of breast cancer may also happen.
 - Other toxicities of the hormonal contraceptives include nausea, breast tenderness, headache, skin pigmentation, and depression.

